Supplemental Material for: Overlapping communities as connected components of a commonality hypergraph

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This Supplemental Material completes the main text. It gives the details of the empirical commonality analysis of Sec. III with the full figures of the four networks; the details of Sec. VII that the main text summarises, with the benchmarks under the label-free protocol (Table S1) and the robustness to noise (Table S2); the benchmarks under the oracle, frozen and native protocols (Tables S3–S5); ONMI and per-community recovery of the oracle and label-free covers (Tables S6–S11); coverage and number of communities of every method (Tables S12–S15); the paired intervals behind the scorecard (Table S16); the label-free selector against the oracle (Table S17); the grid-edge audit (Table S18); the control families (Table S19) and the null graphs without planted communities (Table S20); the commonality family on the benchmarks (Tables S21 and S22); the SNAP graphs detected on their induced subgraphs (Table S23) and scored by F1 against the distinct node sets (Table S24); every method on the fourteen smaller networks (Tables S25–S29); and the numbers quoted in the text that appear in no table, with their sources (Table S30). Figure S1 shows the panels of Fig. 4 under the oracle protocol.

Commonality in empirical networks: details

Unlabelled nodes. Edges touching unlabelled DBLP authors join low-degree nodes in small, tightly knit groups – the median larger endpoint degree of the pooled "different" class is 8, against 22 for same-venue edges – and have high commonality (median $J = 0.40$); pooled into the "different" class they push its median above that of same-venue pairs, and the AUC for separating the classes falls to 0.41 [Fig. S2(f)]. Even after unlabelled edges are set apart, "outer" means no recorded shared group, not proven separation: on the same DBLP co-authorship network with venue groups, and on an Amazon co-purchasing network with product-category groups, standard methods recover the recorded groups poorly [1]. With unlabelled edges set apart, the inner and outer medians of closed-neighbourhood Jaccard are 0.29 and 0.15 in the karate club, 0.25 and 0.17 in DBLP, 0.27 and 0.20 in Amazon, and 0.032 and 0.022 in YouTube [Fig. S2(a–d)].

Degree strata. The stratum AUCs of Amazon and YouTube lie around their pooled 0.60 and 0.61 [Fig. S2(g,h)]. Their weaker signal is consistent with what their ground truth is: Amazon's product categories are nested, with 95% of labelled products in several categories, and YouTube's groups are user-declared interests rather than structural units [2].

The family of functions. With $I = |N[u] \cap N[v]|$ and $U = |N[u] \cup N[v]|$, the eight functions of Fig. S3 are the raw overlap I; Jaccard I/U [3]; the generalisations I^2/U and $I/U^{1/2}$ of the form I^a/U^b (the first ranks pairs as the degree-normalised score $I^2/(\langle d\rangle U)$ of Ref. [4], and the two rank pairs identically, since $I/U^{1/2} = (I^2/U)^{1/2}$, so they count as one choice for any threshold-based use); the cosine $I/\sqrt{|N[u]||N[v]|}$ [5] used by SCAN [6]; the overlap normalised by the smaller and by the larger neighbourhood, $I/\min$ and $I/\max$ (closed-neighbourhood versions of the hub-promoted and hub-depressed indices [7, 8], the first derived from the topological overlap of Ravasz *et al.* [9]); and, as a structurally different score, resource allocation [7] $\sum_{w\in N(u)\cap N(v)} 1/d_w$, which weights shared neighbours by their rarity. The karate club is too small to stratify; there every function separates the classes in aggregate, with AUC 0.73–0.89. That the overlap-based scores perform alike within strata is expected, since the strata are narrow degree bins. Commonality is therefore best understood as a family whose appropriate member depends on the network.

Pairs and triples. We sample wedges whose three nodes are labelled and weight closed triangles by 1/3, so that the population is unique triples rather than centred wedges. The pair rule tests two pairs for an open wedge and three for a triangle; Fig. S4 compares the two rules over the full range of thresholds. At recall 0.2, 91% of pair-admitted triples are pure against 92% of triple-admitted ones in Amazon (base rate 86%), and 10% against 13% in YouTube (base rate 10%). Closed triangles alone are 99% pure in DBLP, reflecting cliques of co-authors, but cover only a quarter of the pure triples; open wedges supply the rest, which is the trade-off the triple rule manages.

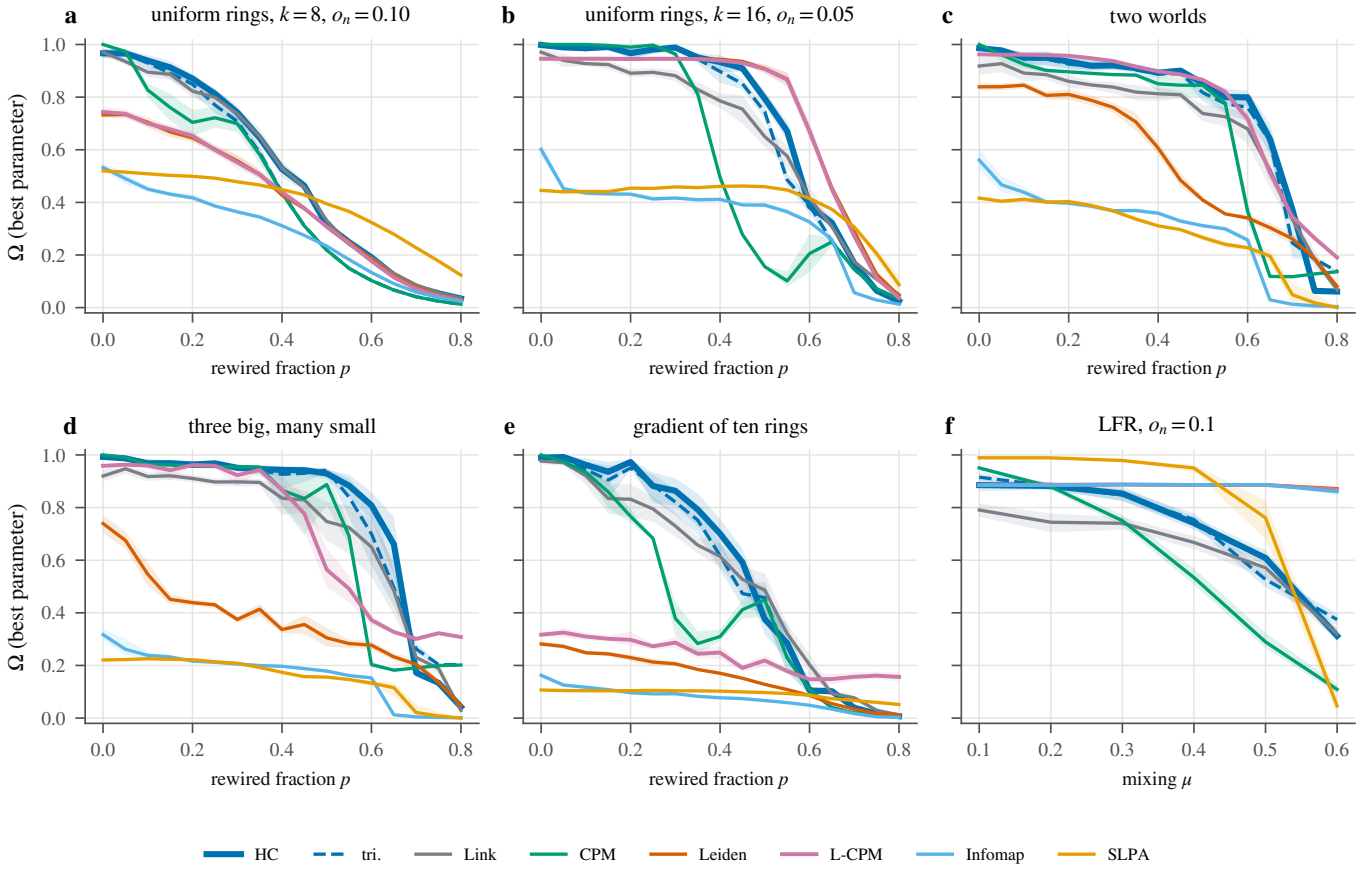


FIG. S1. As Fig. 4, with every method at its best parameter on each graph (oracle protocol).

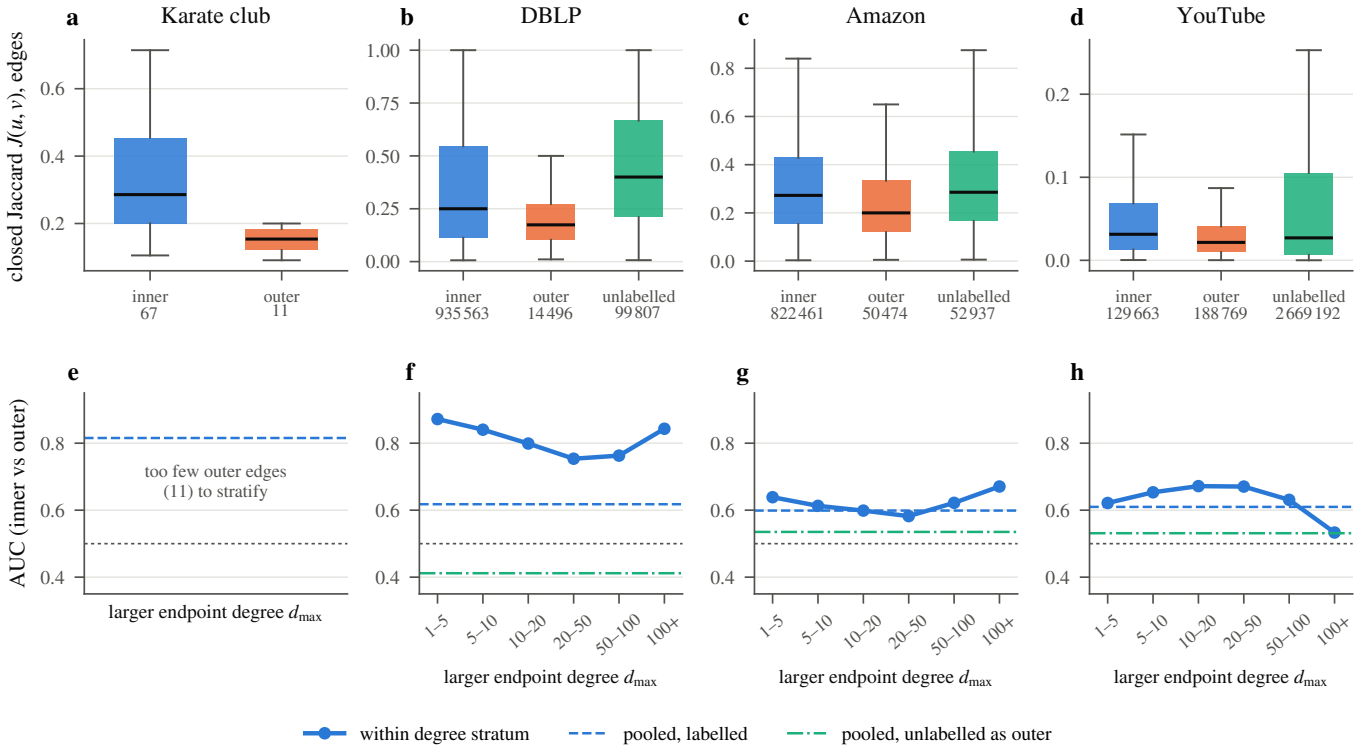


FIG. S2. Commonality of edges in four networks with recorded memberships. (a–d) Closed-neighbourhood Jaccard of edges whose endpoints share a recorded group (inner), are both labelled but share none (outer), or include an unlabelled node; boxes span the quartiles, whiskers the 5th–95th percentiles, numbers give edge counts. (e–h) AUC for separating inner from outer edges within strata of the larger endpoint degree (points), pooled over all labelled edges (dashed), and with unlabelled edges counted as outer (dash-dotted), as in the earlier analysis of Ref. [4].

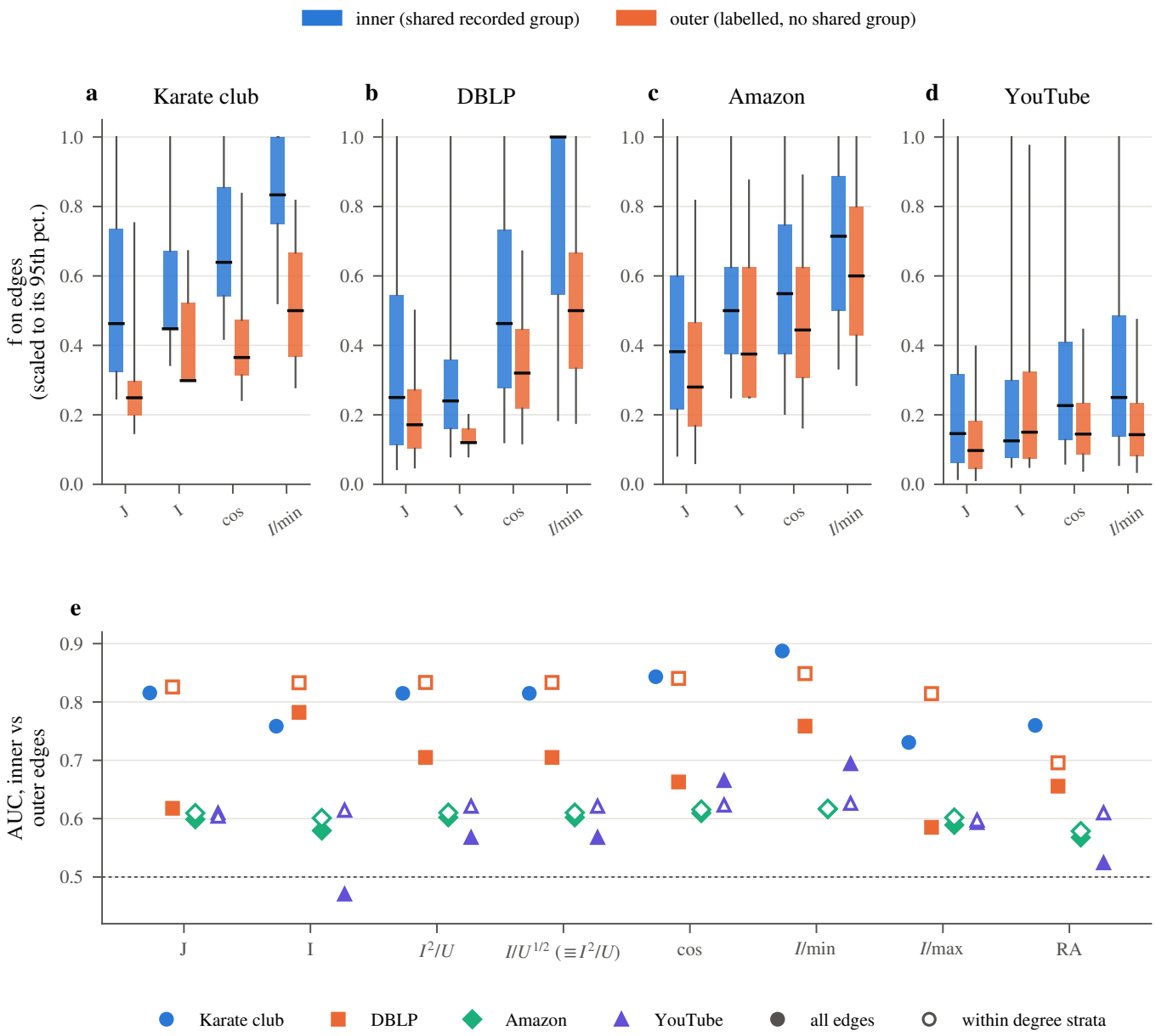


FIG. S3. Commonality as a family of functions. (a–d) Inner (shared recorded group) and outer (labelled, no shared group) edges under four functions, each scaled to its own 95th percentile. (e) AUC for separating inner from outer edges with eight scores, over all labelled edges (filled) and within strata of both endpoint degrees (open); $I/U^{1/2}$ ranks pairs as I^2/U does. The karate club is too small to stratify.

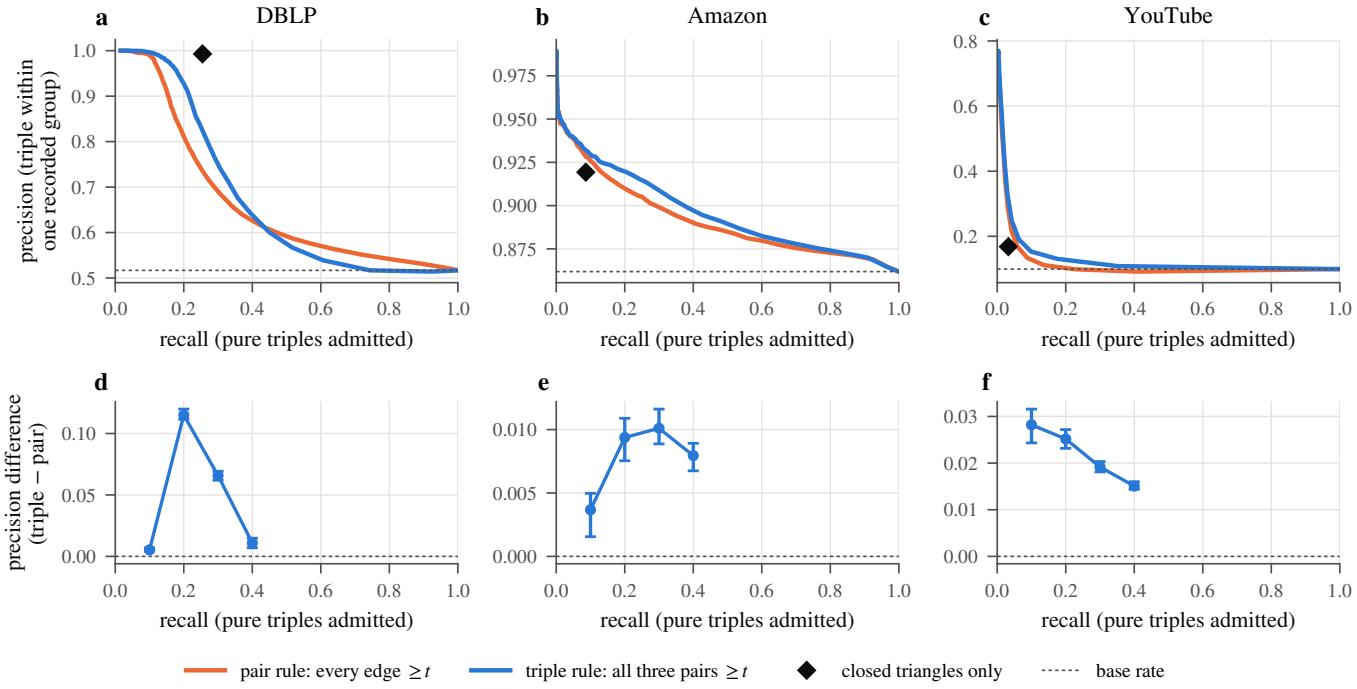


FIG. S4. Pair rule versus triple rule on unique labelled triples (wedges sampled uniformly, triangles weighted 1/3). Precision is the fraction of admitted triples contained in one recorded group, recall the fraction of such triples admitted, as the threshold varies. Diamond: closed triangles only, without a threshold. Bottom row: difference in precision between the triple and the pair rule at equal recall, with 95% intervals from resampling the triples; the y-axes of the top row differ between panels.

Further results on the benchmarks

This section gives the detail behind Secs. VII A–VII G that the main text summarises: the label-free benchmark table (Table S1), the robustness table (Table S2) and the discussion of the remaining comparisons. As in the main text, numbers quoted for a ring family without its overlap level are at $o_n = 0.05$.

Recovery window and fusions. At $o_n = 0.05$, for $k = 16$ the isolation hypothesis of Theorem 2 cannot hold, and for $k = 8$ it would need exactly five merge positions more than k apart on every ring, which random merges essentially never produce. At the thresholds of our grid, rings are then fused inside the window in three ways: two rings share two merged nodes within ring distance k on both rings (Proposition 4); two rings share two merged nodes within $k/2$ on one ring and within $3k/2$ on the other; or the merged nodes of three pairwise-merged rings form an admissible wedge. Without noise and at $o_n = 0.05$, between 6% and 61% of the rings of a family are fused with another ring just above the lower edge of the window, and at $o_n = 0.10$ between 38% and all of them (Table X). For $k = 16$ the noise-free curve of Fig. 3(b) rises from 0.39 at $t = 0.04$ to 0.89 at $t = 0.24$ and reaches 0.95 only for $t \geq 0.30$. At small t, below the lower edge of Theorem 2, communities coalesce through their shared nodes and, with noise, also through rewired edges; as the rewired fraction grows, intra-community commonality erodes and the best threshold moves down, from the window of the noise-free lattice towards $t \approx 0.1$.

Label-free protocol. Table S1 gives the label-free Ω of every method. Of the seven ring settings in which HyperCommon trails without labels, the dense rings with little overlap are the uniform rings with $k = 16$ at $o_n \leq 0.05$ and the dense rings of unequal size without overlap, where Leiden leads. On three large dense rings with sixteen small sparse ones (three big, many small) HyperCommon reaches $\Omega = 0.78$ against 0.68 for clique percolation and 0.34 for Leiden with the Constant Potts Model; on the two-worlds family 0.76 against 0.66 and 0.64; and on two worlds under superlinear preferential rewiring 0.79 against 0.66 for the Constant Potts Model (0.79 at its best resolution; Table S3) and for link communities. On uniform rings HyperCommon leads with $k = 8$ and overlap (0.50 against 0.45 for link communities at $o_n = 0.10$), and modularity-based Leiden leads on dense rings with little overlap (0.84 against 0.78 at $k = 16$, $o_n = 0$), where a single resolution fits every community.

Selector and native protocol. The selector costs the Constant Potts Model more than HyperCommon on unequal rings; Table S17 gives the label-free/oracle ratios and the numbers of communities of the selected covers. Under the native protocol (Table S5) the six ring settings in which HyperCommon is behind are the dense rings with little overlap, where Leiden at $\gamma = 1$ leads, and the gradient, where clique percolation and ANGEL lead; on the eight LFR families it trails, as under every protocol. At the edge density the Constant Potts Model reaches 0.70–0.82 on two worlds at the three overlap levels, against 0.50–0.79 at the resolution of largest modularity. Link communities lose most from their own rule: the cut of maximum partition density returns 640–1320 communities on every ring setting, against 210–930 at the cut of largest modularity, and their Ω falls from 0.39–0.81 to 0.07–0.47.

Oracle protocol. With every method at its best parameter on each graph (Table S3), HyperCommon is ahead, besides three big, many small and the gradient with overlap, on sparse uniform rings with overlap, with or without preferential noise, on sparse rings of unequal size at $o_n = 0.05$ and on dense rings of unequal size at $o_n = 0.10$. Leiden, with one objective or the other, is ahead on the rings without overlap other than three big, many small and the gradient, on the dense rings at $o_n = 0.05$, and on rings of equal size and unequal degree at every overlap level (0.72 against 0.71 at $o_n = 0.05$); the Constant Potts Model, given its best resolution on each graph, is ahead on two worlds at $o_n \leq 0.05$ and under linear preferential noise, by at most 0.05, and level with it at $o_n = 0.10$ and under superlinear noise; link communities lead on the gradient without overlap. On the gradient at $p = 0$, the resolution at which Leiden returns the right number of communities (9.9 on average against 10) gives $\Omega = 0.26$, and HyperCommon attains 0.97 at the threshold at which it returns ten communities.

The gradient. The label-free cover on the gradient, too fine because the selector picks a threshold above the oracle's on 113 of the 170 test graphs, has lower coverage than the oracle cover (0.91 against 0.96), and the large and small rings are both recovered worse (0.26 and 0.31 per community, against 0.49 and 0.48).

Per-community recovery and ONMI. On the gradient HyperCommon recovers the large rings best at every overlap level and the small rings best at $o_n \leq 0.05$ (at $o_n = 0.10$ DEMON recovers them slightly better, 0.36 against 0.35; Tables S10 and S11). On the two families with a few large dense rings and many small sparse ones it recovers the large rings best (0.84–0.85 against 0.72–0.83 for the Constant Potts Model) and the Constant Potts Model recovers the small rings best (0.53–0.54 against 0.33–0.38): at the best threshold for the large rings, part of each small sparse ring has commonality below it, and HyperCommon splits such rings into pieces and leaves part of their nodes unassigned (it covers 87%–88% of the nodes on these two families at its best threshold; the coverage and number of communities of every method are in Tables S12 to S15). Scored by ONMI on the same covers (Table S6) and by recovery averaged over all communities (Table S7), the ranking changes on these two families. The Constant Potts Model, already slightly ahead of HyperCommon by Ω on two worlds (0.783 against 0.775), is clearly ahead there (ONMI 0.57 against 0.42; all-community recovery 0.60 against 0.48), and it is also ahead on three big, many small (0.52 against 0.40; 0.56 against 0.41), where HyperCommon leads by Ω, because it recovers the small sparse rings better; by ONMI it

TABLE S1. Accuracy on the benchmarks as a user would run the methods, every parameter chosen without labels by one rule: on each test graph, the parameter whose cover has the largest overlapping modularity, Eq. (7), which for a partition is the Newman–Girvan modularity (label-free protocol). Ω, mean over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level. HC: HyperCommon; e-c.: its triples joined only through shared edges; tri.: triangles only (J-thresholded 3-clique percolation [10, 11]); 2-test: third pair not tested; Link: link communities with the same similarity; CPM: clique percolation; Leiden: Leiden with the Reichardt–Bornholdt modularity; L-CPM: Leiden with the Constant Potts Model; Info.: Infomap; Walk.: Walktrap, walks of length 4, dendrogram cut at maximum modularity; LPA: label propagation; SBM: nested degree-corrected stochastic block model, each level of its hierarchy a parameter value; SCAN: SCAN with cosine similarity; SCAN-J: SCAN with Jaccard similarity; α: exponent of the preferential (degree-biased) noise. Parameter-free methods (Walktrap, LPA) are as under every protocol. Superscripts on HC: the paired 95% bootstrap interval of its difference from the best other method (ablations excluded) lies above zero (+), contains zero (=) or lies below zero (−). Bold: best in row, within 0.005. 95% bootstrap half-widths of these means over the ten test seeds have median 0.007 and maximum 0.051.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	**0.61**$^{=}$	**0.61**	0.51	0.51	0.60	0.51	**0.61**	**0.61**	0.48	0.39	0.14	0.38	0.36	0.53	0.52	0.50	0.53
	0.05	**0.55**$^{+}$	0.54	0.49	0.29	0.53	0.49	0.44	0.43	0.32	0.24	0.12	0.26	0.33	0.44	0.39	0.26	0.30
	0.10	**0.50**$^{+}$	**0.50**	0.46	0.22	0.45	0.44	0.32	0.31	0.26	0.18	0.11	0.21	0.30	0.40	0.40	0.21	0.21
uniform, k=16	0.00	0.78^{-}	0.78	0.75	0.69	0.76	0.67	**0.84**	0.84	0.47	0.65	0.14	0.50	0.34	0.62	0.67	0.65	0.59
	0.05	0.72^{-}	0.71	0.69	0.39	0.54	0.56	**0.75**	0.75	0.34	0.47	0.11	0.38	0.32	0.50	0.63	0.33	0.35
	0.10	**0.62**$^{=}$	**0.62**	0.60	0.29	0.39	0.54	**0.62**	0.61	0.29	0.33	0.09	0.29	0.29	0.45	0.57	0.24	0.25
equal k, unequal s, k=8	0.00	**0.57**$^{+}$	0.57	0.46	0.49	0.56	0.46	0.56	0.56	0.36	0.33	0.10	0.33	0.25	0.48	0.49	0.47	0.49
	0.05	**0.52**$^{+}$	0.51	0.45	0.25	0.49	0.43	0.35	0.34	0.24	0.19	0.09	0.23	0.23	0.38	0.35	0.22	0.23
	0.10	**0.45**$^{+}$	**0.45**	0.40	0.17	0.41	0.37	0.24	0.24	0.19	0.15	0.08	0.16	0.21	0.33	0.33	0.15	0.16
equal k, unequal s, k=16	0.00	0.80^{-}	0.79	0.77	0.70	0.77	0.71	**0.84**	**0.84**	0.61	0.69	0.17	0.49	0.43	0.64	0.69	0.66	0.60
	0.05	**0.73**$^{=}$	**0.73**	0.71	0.42	0.56	0.59	**0.73**	0.72	0.50	0.56	0.14	0.41	0.41	0.51	0.66	0.37	0.37
	0.10	**0.65**$^{+}$	**0.65**	0.62	0.31	0.42	0.57	0.60	0.59	0.44	0.44	0.12	0.33	0.37	0.45	0.61	0.27	0.29
equal s, unequal k	0.00	0.78$^{=}$	**0.78**	0.76	0.71	0.74	0.60	**0.79**	**0.79**	0.55	0.60	0.16	0.50	0.35	0.53	0.64	0.66	0.61
	0.05	0.70$^{=}$	**0.70**	0.69	0.47	0.57	0.59	0.67	**0.70**	0.45	0.52	0.13	0.40	0.31	0.38	0.57	0.40	0.41
	0.10	0.61^{-}	0.61	0.59	0.39	0.45	0.56	0.58	**0.63**	0.39	0.46	0.11	0.32	0.27	0.29	0.46	0.32	0.34
two worlds	0.00	**0.83**$^{+}$	**0.82**	0.82	0.71	0.78	0.70	0.58	0.79	0.40	0.52	0.13	0.28	0.28	0.48	0.65	0.72	0.64
	0.05	**0.76**$^{+}$	**0.76**	**0.76**	0.58	0.66	0.66	0.44	0.64	0.29	0.41	0.10	0.22	0.25	0.35	0.55	0.51	0.49
	0.10	**0.71**$^{+}$	**0.71**	0.70	0.49	0.53	0.65	0.38	0.50	0.26	0.32	0.08	0.18	0.22	0.28	0.43	0.42	0.43
three big, many small	0.00	**0.83**$^{+}$	**0.83**	0.82	0.76	0.81	0.72	0.33	0.40	0.21	0.27	0.07	0.19	0.15	0.47	0.64	0.74	0.66
	0.05	**0.78**$^{+}$	**0.78**	0.74	0.48	0.67	0.68	0.28	0.34	0.16	0.22	0.06	0.13	0.14	0.35	0.51	0.42	0.44
	0.10	**0.67**$^{+}$	0.66	0.62	0.36	0.55	0.61	0.23	0.29	0.14	0.17	0.05	0.09	0.12	0.29	0.43	0.33	0.40
gradient	0.00	0.42^{-}	0.40	0.39	0.36	**0.61**	0.56	0.21	0.21	0.11	0.14	0.03	0.14	0.07	0.50	0.48	0.41	0.38
	0.05	0.32^{-}	0.31	0.27	0.20	**0.48**	0.40	0.14	0.15	0.07	0.10	0.03	0.08	0.07	0.34	0.42	0.18	0.16
	0.10	0.23^{-}	0.24	0.22	0.13	**0.39**	0.35	0.09	0.11	0.06	0.07	0.02	0.05	0.06	0.27	0.33	0.12	0.13
uniform k=8, α=1	0.05	**0.56**$^{+}$	0.55	0.49	0.31	0.53	0.48	0.45	0.44	0.32	0.24	0.12	0.28	0.33	0.43	0.37	0.28	0.32
uniform k=8, α=2	0.05	**0.56**$^{+}$	**0.56**	0.49	0.33	0.54	0.48	0.45	0.45	0.33	0.22	0.12	0.27	0.33	0.43	0.39	0.30	0.32
two worlds, α=1	0.05	**0.77**$^{+}$	**0.77**	0.75	0.54	0.66	0.65	0.47	0.66	0.31	0.62	0.11	0.26	0.26	0.26	0.44	0.46	0.47
two worlds, α=2	0.05	**0.79**$^{+}$	**0.79**	0.75	0.51	0.66	0.64	0.49	0.66	0.32	0.62	0.10	0.29	0.26	0.23	0.41	0.50	0.50
LFR, o_n=0		0.84^{-}	0.86	0.84	0.56	0.85	0.66	0.98	0.98	**1.00**	0.98	0.61	0.99	0.91	0.46	0.77	0.43	0.39
LFR, o_n=0.1		0.69^{-}	0.72	0.71	0.27	0.53	0.60	0.78	0.78	**0.88**	0.81	0.49	0.87	0.79	0.41	0.72	0.29	0.23
LFR, o_n=0.3		0.45^{-}	0.50	0.46	0.11	0.31	0.48	0.54	0.53	**0.64**	0.49	0.32	0.47	0.60	0.40	0.64	0.13	0.10
LFR, $\langle k \rangle$=10		0.43^{-}	0.45	0.35	0.09	0.40	0.36	0.69	0.70	**0.71**	0.61	0.36	0.63	0.48	0.26	0.25	0.12	0.12
LFR, o_m=4		0.61^{-}	0.63	0.62	0.40	0.46	0.52	0.62	0.62	**0.71**	0.62	0.40	0.67	0.68	0.43	0.66	0.36	0.28
LFR, sizes 50–250		0.42^{-}	0.41	0.42	0.14	0.40	0.16	**0.87**	0.85	0.77	0.75	0.38	0.79	0.59	0.18	0.55	0.13	0.07
LFR, C_{LFR}=0.3		0.33^{-}	0.33	0.32	0.19	0.17	0.39	0.50	0.50	0.55	0.52	0.48	0.39	**0.56**	0.18	0.30	0.20	0.20
LFR, C_{LFR}=0.5		0.24^{-}	0.24	0.24	0.19	0.10	0.24	0.29	0.29	0.30	0.26	**0.35**	0.23	**0.35**	0.17	0.21	0.18	0.20

is also ahead on two worlds with linear preferential noise (0.58 against 0.51). HyperCommon stays ahead by ONMI on the gradient, on sparse uniform rings with overlap, on sparse rings of unequal size with overlap and on two worlds with superlinear preferential noise; over the 28 ring settings it leads by ONMI on 10 and is level on 3, and by all-community recovery it leads on 9 and is level on 3. On the label-free covers (Tables S8 and S9) the same metrics put HyperCommon ahead on 15 settings by ONMI (level on 1) and on 12 by all-community recovery (level on 5). Chosen by modularity, the resolution of the Constant Potts Model recovers the small rings less well, and at $o_n = 0.05$ its lead by these metrics on the two unequal families is gone (ONMI 0.45 against 0.45 on two worlds and 0.27 against 0.39 on three big, many small; all-community recovery within 0.01 on both); it keeps a lead in all-community recovery on

TABLE S2. Robustness to noise on the ring families: mean Ω over the high-noise half ($p \in [0.4, 0.8]$) and $p_{1/2}$, the rewired fraction at which the mean Ω over the ten test graphs first falls below 0.5 (linear interpolation; > 0.8: never), for HyperCommon (HC) and the best of the other methods by each measure, under the oracle and label-free protocols. Bold: HC ahead or within 0.005 (0.01 for $p_{1/2}$).

	oracle				label-free			
Family	HC $\bar{\Omega}_{\text{hi}}$	best other	HC $p_{1/2}$	best other	HC $\bar{\Omega}_{\text{hi}}$	best other	HC $p_{1/2}$	best other
uniform, k=8, o_n=0	0.29	0.39 (SLPA)	0.48	0.52 (Leiden)	0.28	0.30 (SLPA)	**0.48**	0.48 (L-CPM)
uniform, k=8, o_n=0.05	0.25	0.35 (SLPA)	**0.46**	0.45 (Link)	0.24	0.27 (SLPA)	**0.45**	0.45 (Link)
uniform, k=8, o_n=0.10	0.22	0.31 (SLPA)	**0.42**	0.42 (Link)	0.21	0.24 (SLPA)	**0.41**	0.38 (Link)
uniform, k=16, o_n=0	0.56	0.72 (Leiden)	0.61	0.68 (Leiden)	0.56	0.69 (Leiden)	0.61	0.67 (Leiden)
uniform, k=16, o_n=0.05	0.47	0.59 (Leiden)	0.58	0.64 (Leiden)	0.46	0.56 (Leiden)	0.58	0.63 (Leiden)
uniform, k=16, o_n=0.10	0.37	0.44 (Leiden)	0.54	0.58 (Leiden)	0.36	0.40 (Leiden)	0.53	0.57 (Leiden)
equal k, unequal s (k=8)	0.19	0.26 (SLPA)	**0.41**	0.40 (Link)	**0.19**	0.19 (SLPA)	**0.41**	0.40 (Link)
equal k, unequal s (k=16)	0.51	0.63 (Leiden)	0.60	0.68 (Leiden)	0.49	0.54 (Leiden)	0.59	0.64 (Leiden)
equal s, unequal k	0.53	0.60 (L-CPM)	0.66	0.69 (L-CPM)	0.52	0.55 (L-CPM)	**0.66**	0.66 (L-CPM)
two worlds	0.61	0.62 (L-CPM)	**0.68**	0.66 (Link)	**0.59**	0.52 (Link)	**0.68**	0.65 (Link)
three big, many small	**0.63**	0.54 (Link)	**0.67**	0.65 (Link)	**0.60**	0.49 (Link)	**0.66**	0.62 (Link)
gradient of ten rings	0.23	0.26 (Link)	0.47	0.48 (Link)	0.14	0.18 (Link)	0.17	0.41 (Link)
uniform k=8, pref. α=1	0.25	0.35 (SLPA)	**0.46**	0.46 (Leiden)	0.24	0.27 (SLPA)	**0.46**	0.45 (Link)
uniform k=8, pref. α=2	0.25	0.34 (SLPA)	**0.46**	0.46 (Leiden)	0.24	0.27 (SLPA)	**0.46**	0.45 (Link)
two worlds, pref. α=1	0.60	0.70 (L-CPM)	0.67	0.70 (L-CPM)	**0.59**	0.50 (L-CPM)	0.67	0.69 (L-CPM)
two worlds, pref. α=2	0.61	0.64 (L-CPM)	**0.67**	0.66 (L-CPM)	**0.60**	0.48 (L-CPM)	**0.67**	0.62 (L-CPM)

two worlds without overlap (0.70 against 0.67) and on both families at $o_n = 0.10$ (0.35 against 0.33 and 0.30 against 0.25).

Overlap level. On two worlds HyperCommon and the Constant Potts Model both fall from 0.84 to 0.72 with labels as the overlap grows (the Constant Potts Model significantly ahead at $o_n \leq 0.05$); the steeper label-free decline of the Constant Potts Model there (0.79 to 0.50, against 0.83 to 0.71 for HyperCommon) comes from the selection of its resolution.

Robustness to noise

The planted structure is erased as $p \to 1$, and every method's accuracy falls with it; the question is how long each holds on. Table S2 reports, for every ring family, the rewired fraction $p_{1/2}$ at which the mean Ω first falls below one half and the mean Ω over the high-noise half, $p \in [0.4, 0.8]$. Without labels, on the four families whose large rings have degree 32 and small rings degree 8 (two worlds under three noise models, and three big, many small) HyperCommon has the highest mean Ω over the high-noise half: 0.60 against 0.49 for link communities on three big, many small, 0.59 against 0.52 on two worlds [Fig. 4(c,d)], and 0.60 against 0.48 for the Constant Potts Model under superlinear preferential rewiring; its accuracy falls below that of the partitions only near the end of the range, from $p = 0.70$ on three big, many small and $p = 0.75$ on two worlds. With labels, its $p_{1/2}$ is the latest or within 0.01 of it in eight families, and its high-noise mean Ω is the highest only on three big, many small (0.63 against 0.54); on two worlds under the three noise models the Constant Potts Model is higher (0.62–0.70 against 0.60–0.61). On the gradient, without labels, it is the least robust of the leading methods (0.14 against 0.18 for link communities; $p_{1/2}$ 0.17 against 0.41). Of the other methods, only the ego-network methods DEMON and ANGEL degrade under degree-biased noise; for HyperCommon the bound on Jaccard commonality by the degree ratio of a pair offers a mechanism consistent with its robustness: a hub grown by preferential rewiring has low commonality with the lattice neighbours it attaches to, so the edges it gains rarely pass. On uniform sparse rings at high noise, SLPA's slowly decaying cover has the highest Ω, and on uniform dense rings Leiden does.

Ablations and the commonality function

Ablations. Without the third-pair test, a wedge whose two edges pass reaches into another community through a shared node or through an edge between communities whose own commonality passes the threshold, which is how

the two-test variant loses 0.21 even on LFR graphs without overlap. Triangles alone are better than HyperCommon in one setting under the oracle protocol, and higher in mean in three, by at most 0.02 (heavily overlapping and large-community LFR graphs), and in none under the label-free one, where the largest difference in their favour, 0.02, is within the interval. Joining triples only through shared edges is slightly better on the LFR graphs, by up to 0.03 with labels and 0.05 without, most at $o_n = 0.3$, with the exception of the large-community slice without labels and by less than 0.01 on the clustered families. With labels, link communities are more accurate than HyperCommon on the LFR graphs with mean degree 10 and with large communities.

The commonality function. Tables S21 and S22 run HyperCommon with each member of the commonality family on eighteen benchmark settings (the six uniform-ring settings, the six unequal-ring families at $o_n = 0.05$ and the six LFR families without added clustering), the threshold taken as a quantile of each function's own triple scores, so that the functions are compared on matched quantile grids; triples whose score ties with the threshold are all admitted, so the admitted fraction can differ from the nominal level when scores take few values. With the best threshold per graph, Jaccard is the best function, or within 0.01 of it at the precision of Table S21, on seventeen of the eighteen settings, and never more than 0.02 behind (three big, many small: 0.81 against 0.83); no other function is significantly ahead of it on more than five. Where another function leads, the lead is small and follows the structure: the cosine is ahead on four of the six LFR families, by 0.01, and the overlap normalised by the larger neighbourhood on the rings of unequal size, by up to 0.02 on three big and many small rings. The two scores that are not normalised by the neighbourhoods, the raw overlap and resource allocation, are less accurate than Jaccard on all eighteen settings, by up to 0.21 and 0.20: on graphs whose communities differ in degree, a threshold on an unnormalised score is a threshold on degree as much as on commonality, as on the real networks (Sec. III). Choosing the function together with the threshold on each graph raises Ω by 0.01–0.06, most on the gradient of ten rings (0.59 to 0.65) and on LFR with large communities (0.45 to 0.51).

Under the label-free protocol the ordering is much the same, with one exception that bears on the threshold selection. On the gradient of ten rings, the family on which overlapping modularity selects too fine a cover, the selected cover of I^2/U reaches $\Omega = 0.50$ against 0.36 for Jaccard; the same selector, asked to choose the function as well, does not find it (0.31). The function and the threshold are therefore best chosen with the same care, and a selection criterion that does not favour fragmented covers would gain on both.

LFR graphs

With its best threshold HyperCommon reaches $\Omega = 0.885$ at $\mu = 0.1$ and $o_n = 0.1$, against 0.884–0.886 for Leiden and Infomap, while SLPA reaches 0.99 [Fig. S1(f)]. LFR communities are random graphs internally, with degree independent of size; at $\mu = 0.3$ only 5% of the wedges inside communities pass at $t = 0.1$ in the slice with communities of 50 to 250 nodes. The generator's own rewiring, used to raise the clustering, adds triangles inside communities but also across them, where they were rare: at $\mu = 0.3$ the share of triangles that span two communities grows from 2% to 19% and 32% at the two clustering levels. A second difference from the rings is the overlap itself: an overlapping LFR node splits its degree among its communities, while a merged ring node keeps both neighbourhoods; the split-overlap control below isolates this.

Lattice and random communities, controls and null graphs

Geometry families. Two adjacent members of a random community of the same size and mean degree as a ring share about $(k-1)^2/s$ neighbours, one on the uniform rings. On the noiseless graphs the mean clustering falls from 0.62–0.68 at $q = 0$ to 0.48–0.51 at $q = 0.1$, 0.25–0.30 at $q = 0.3$ and 0.04–0.16 at $q = 1$ (Table II), and the number of triangles per edge on the uniform rings from 4.5 to 1.2. On lattice communities HyperCommon leads with labels by 0.94 against 0.87 for clique percolation on three big, many small and 0.74 against 0.70 for link communities on the gradient, and is level with the Constant Potts Model on two worlds (0.91 for both); without labels it is ahead with 0.89 against 0.82 on two worlds and 0.93 against 0.85 on three big, many small, while on the gradient overlapping modularity again selects too fine a cover (0.42 against 0.64 for link communities). Once a tenth of each community's edges are rewired internally, the leaders with labels are Leiden with modularity on the uniform rings and with the Constant Potts Model on the other three families. From $q = 0.3$ on, the best other method is ahead by 0.11–0.42 with labels at $q = 0.3$; the leaders are Leiden with either objective with labels, and without labels the Constant Potts Model, Infomap on the uniform rings and Walktrap on the gradient at $q = 0.3$. Under the oracle protocol, at $q = 1$ HyperCommon returns about twice as many communities as at $q = 0$ on the uniform rings (208 against 109, for 40 planted) and covers 71% of the nodes instead of 90% on three big, many small. With degree-preserving double-edge

swaps inside each ring instead of endpoint moves (Table S19), the Ω of the leading method is within 0.03 of its value under the endpoint rule at $q = 0.1$ and 0.3, on the uniform rings and on three big, many small, under both protocols.

Open wedges and triangles. On two worlds at $q = 1$ HyperCommon and triangles alone are level in Ω with labels (0.66), but HyperCommon's per-community score is 0.62 against 0.37. The one family where triangles alone are significantly ahead in Ω with labels is three big, many small at $q \geq 0.6$ (0.54 against 0.28 at $q = 1$); without labels they are also higher on two worlds at $q = 1$ (0.66 against 0.62, not significantly). On three big, many small Ω is dominated by the three large rings (Sec. VI E), which triangles alone recover better (0.58 against 0.32 per community), while they recover the small rings worse (0.28 against 0.47), and both their per-community score (0.33 against 0.45) and their ONMI (0.15 against 0.22) are lower.

Controls. Three further controls change one assumption of the ring benchmark at a time (Table S19). With hubs inside the communities, HyperCommon's Ω moves by at most 0.03 from that of the uniform rings and of two worlds, and without labels it stays ahead on the sparse rings (0.56 against 0.53 for link communities); the Leiden objectives, however, gain with hubs and are ahead under both protocols, Leiden on the dense hub rings (0.83 against 0.69 without labels) and the Constant Potts Model on two worlds with hubs (0.83 against 0.75). With split overlap, in which a shared node divides its edges between its rings as in LFR graphs, HyperCommon's Ω is within 0.05 of that of the merged-node families, mostly higher, and it is level with the best other method or ahead on two worlds and on three big, many small under both protocols; the way LFR graphs share nodes is therefore not what limits it there. With random geometric communities, clustered but not lattices, HyperCommon is behind Leiden with labels and level with it without labels on the uniform rings (0.46 against 0.46), and behind the Constant Potts Model on two worlds (0.73 against 0.79 without labels), consistent with the role of the lattice geometry.

Null graphs. On graphs with triadic closure but no planted communities (Table S20), label-free HyperCommon returns 101 communities on the random geometric graph and 175 on the Holme–Kim graph, covering 98% and 96% of the nodes, against 31–36 and 12–27 for Leiden, Infomap and the stochastic block model, whose finest level has 132 and 61 groups.

Real networks

SNAP graphs. At the resolution of largest overlapping modularity, Leiden's F1 on Amazon, DBLP and YouTube (0.53, 0.29 and 0.19) is about half of its best values (0.92, 0.54, 0.36) and the same as at its default resolution $\gamma = 1$ (Table III). With labels, Infomap leads by F1 on Amazon and DBLP and Leiden on YouTube. By Ω the label-free choice costs HyperCommon nothing on Amazon and DBLP, while on YouTube both covers are near chance (-0.007 against 0.017). Infomap at its default Markov time is more accurate on Amazon and DBLP (0.92 and 0.55) than every method under the overlapping-modularity selector, 0.06–0.07 ahead of label-free HyperCommon. By Ω, scored against the distinct ground-truth node sets (the top-5000 lists repeat some sets; Table VI), the modularity- and flow-based partitions lead on all three, and HyperCommon covers 99%, 94% and 67% of the scored nodes at its label-free threshold. Open wedges and the third pair behave mostly as on the benchmarks, with small exceptions: on Amazon, whose co-purchase communities are sparse in triangles, the two-test variant reaches F1 0.887 (but a lower Ω, 0.424 against 0.459), and triangles alone score slightly higher on YouTube without labels (F1 0.300 against 0.289; Ω 0.006 against -0.007) and on DBLP by Ω with labels (0.030 against 0.024). On the subgraphs induced by the labelled nodes (Table S23) the connected components, returned as communities, score F1 0.958 and 0.504 on Amazon and YouTube, within 0.005 of the best method, while Ω still separates them (on YouTube 0.001 for the components against 0.159 for Leiden). On the induced DBLP graph HyperCommon's F1 is 0.584 with and without labels; without labels it is second to SLPA (0.604), and with labels Leiden, SLPA and Infomap reach 0.59–0.61 and HC_f (the best of the seven commonality functions) 0.61. Clique percolation did not complete on the induced YouTube graph, exhausting more than 100 GB of memory.

Fourteen smaller networks. HC_f in Table IV chooses, with labels, among 700 (function, threshold) pairs per network; it is the most accurate method on four networks (college football and three ego networks, by 0.02–0.04 in F1; the same four with 30 quantile levels per function), and chosen without labels it is the most accurate on none by F1 (level on Facebook ego 414) and on two by Ω (college football and political books). With the best threshold the two-test variant has the higher F1 on nine of the fourteen networks, seven of them Facebook ego networks, by 0.005–0.05 (Table S25). Label-free HyperCommon with Jaccard is the most accurate method on none of the fourteen by F1 and on one by Ω (Facebook ego 414); SLPA has the best mean rank without labels (6.0), and SCAN and Leiden with labels (5.4 and 5.5). The best member of the family differs from network to network: resource allocation, normalisation by the smaller neighbourhood, I^2/U and the cosine are each the best on at least one network. On the remaining networks the most accurate method itself varies – SLPA, Leiden, label propagation, SCAN and Infomap each lead on some. By Ω (Table V) HC_f is again the most accurate method on four networks (college football, political books and two ego networks), and the ranking on the others is much as by F1, with one exception: on political blogs

the giant component, which F1 credits as a match for one of the two groups, has Ω near zero (0.04, against 0.55 for Leiden).

Running time

On rings of 190 000 nodes with $k = 16$ (2.7×10^7 wedges) the one-off scoring of the wedges peaks at 5.8 GB of traced memory, and the label-free sweep over 30 thresholds, with the overlapping modularity of every cover, adds 73 s to the 45 s of the scoring (Fig. 5). On the full Amazon and DBLP graphs, timed as in Sec. VII G, one run of label propagation takes 19 and 31 s. The local query is fast at every size because it reads only the neighbourhoods of the community it grows and of its frontier.

TABLE S3. As Table S1, with every method at its best parameter on each test graph (oracle protocol). 95% bootstrap half-widths of these means over the ten test seeds have median 0.005 and maximum 0.049. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.62$^-$	0.62	0.51	0.55	0.62	0.51	**0.68**	**0.67**	0.49	0.39	0.14	0.41	0.47	0.54	0.54	0.54	0.54
	0.05	**0.57**$^+$	**0.57**	0.50	0.34	**0.57**	0.49	0.55	0.55	0.35	0.24	0.12	0.31	0.44	0.44	0.43	0.30	0.31
	0.10	**0.52**$^+$	0.52	0.47	0.25	0.52	0.45	0.41	0.41	0.28	0.18	0.11	0.24	0.40	0.40	0.41	0.22	0.22
uniform, k=16	0.00	0.78$^-$	0.78	0.75	0.70	0.78	0.68	**0.86**	0.85	0.47	0.65	0.14	0.50	0.44	0.64	0.67	0.65	0.59
	0.05	0.73$^-$	0.72	0.70	0.42	0.66	0.56	**0.77**	**0.77**	0.35	0.47	0.11	0.38	0.41	0.51	0.63	0.34	0.36
	0.10	0.63$^=$	0.63	0.61	0.31	0.53	0.54	**0.64**	0.63	0.29	0.33	0.09	0.29	0.38	0.45	0.57	0.24	0.26
equal k, unequal s, k=8	0.00	0.58$^-$	0.57	0.46	0.51	0.57	0.46	**0.60**	**0.60**	0.36	0.33	0.10	0.34	0.35	0.49	0.49	0.49	0.49
	0.05	**0.52**$^+$	0.52	0.45	0.27	0.52	0.43	0.40	0.39	0.25	0.19	0.09	0.24	0.32	0.39	0.35	0.23	0.24
	0.10	**0.46**$^=$	0.45	0.41	0.19	**0.46**	0.37	0.29	0.29	0.20	0.15	0.08	0.18	0.29	0.35	0.34	0.16	0.17
equal k, unequal s, k=16	0.00	0.80$^-$	0.79	0.77	0.72	0.79	0.71	**0.87**	**0.87**	0.61	0.69	0.17	0.50	0.53	0.67	0.70	0.66	0.60
	0.05	0.75$^-$	0.74	0.72	0.47	0.69	0.59	**0.79**	0.78	0.51	0.56	0.14	0.43	0.50	0.55	0.66	0.39	0.39
	0.10	**0.67**$^+$	**0.67**	0.65	0.35	0.59	0.57	0.66	0.65	0.44	0.44	0.12	0.35	0.46	0.49	0.62	0.28	0.29
equal s, unequal k	0.00	0.79$^-$	0.79	0.77	0.73	0.78	0.61	**0.82**	**0.82**	0.55	0.60	0.16	0.50	0.40	0.54	0.64	0.68	0.61
	0.05	0.71$^-$	0.72	0.70	0.49	0.64	0.59	**0.72**	**0.72**	0.45	0.52	0.13	0.42	0.35	0.41	0.58	0.41	0.42
	0.10	0.62$^-$	0.62	0.60	0.41	0.53	0.56	0.63	**0.65**	0.39	0.46	0.11	0.34	0.31	0.34	0.49	0.32	0.35
two worlds	0.00	**0.84**$^-$	**0.83**	0.82	0.75	0.82	0.70	0.67	**0.84**	0.40	0.52	0.13	0.30	0.32	0.53	0.66	0.72	0.64
	0.05	0.78$^-$	0.78	0.77	0.59	0.71	0.68	0.56	**0.78**	0.29	0.41	0.10	0.24	0.28	0.40	0.58	0.51	0.49
	0.10	**0.72**$^=$	**0.72**	0.71	0.50	0.61	0.66	0.45	**0.72**	0.26	0.32	0.08	0.20	0.24	0.35	0.49	0.43	0.44
three big, many small	0.00	**0.84**$^+$	**0.84**	0.84	0.77	0.83	0.74	0.49	0.81	0.24	0.27	0.07	0.25	0.18	0.50	0.65	0.75	0.67
	0.05	**0.80**$^+$	**0.80**	0.79	0.57	0.72	0.71	0.37	0.71	0.16	0.22	0.06	0.16	0.16	0.38	0.53	0.49	0.50
	0.10	0.71$^+$	**0.72**	0.69	0.48	0.61	0.67	0.26	0.48	0.14	0.17	0.05	0.12	0.13	0.33	0.45	0.42	0.44
gradient	0.00	0.65$^-$	0.64	0.60	0.57	**0.66**	0.57	0.35	0.44	0.14	0.14	0.03	0.17	0.11	0.54	0.54	0.54	0.51
	0.05	**0.57**$^+$	**0.57**	0.54	0.29	0.54	0.43	0.16	0.23	0.07	0.10	0.03	0.08	0.09	0.43	0.47	0.22	0.23
	0.10	**0.48**$^+$	**0.48**	0.47	0.25	0.45	0.41	0.10	0.18	0.06	0.07	0.02	0.06	0.08	0.37	0.37	0.16	0.18
uniform k=8, α=1	0.05	**0.58**$^+$	0.57	0.50	0.35	0.57	0.49	0.56	0.55	0.35	0.24	0.12	0.32	0.44	0.44	0.43	0.31	0.33
uniform k=8, α=2	0.05	**0.58**$^+$	**0.58**	0.50	0.38	0.57	0.48	0.56	0.55	0.35	0.22	0.12	0.32	0.44	0.44	0.43	0.33	0.36
two worlds, α=1	0.05	0.78$^-$	0.78	0.76	0.56	0.70	0.67	0.61	**0.82**	0.31	0.62	0.11	0.27	0.30	0.35	0.47	0.47	0.47
two worlds, α=2	0.05	**0.79**$^=$	**0.79**	0.77	0.57	0.72	0.65	0.66	**0.79**	0.34	0.62	0.10	0.29	0.29	0.32	0.44	0.51	0.52
LFR, o_n=0		0.86$^-$	0.86	0.85	0.65	0.85	0.66	**1.00**	**1.00**	**1.00**	0.98	0.61	0.99	0.91	0.51	0.77	0.57	0.41
LFR, o_n=0.1		0.74$^-$	0.75	0.73	0.34	0.66	0.60	**0.88**	**0.88**	**0.88**	0.81	0.49	0.87	0.84	0.46	0.73	0.33	0.25
LFR, o_n=0.3		0.50$^-$	0.53	0.51	0.16	0.42	0.48	0.69	0.69	0.65	0.49	0.32	0.57	**0.71**	0.42	0.65	0.15	0.11
LFR, $\langle k \rangle$=10		0.46$^-$	0.47	0.36	0.12	0.51	0.36	**0.79**	**0.79**	0.71	0.61	0.36	0.63	0.53	0.27	0.31	0.15	0.13
LFR, o_m=4		0.62$^-$	0.63	0.62	0.44	0.60	0.52	0.71	0.71	0.71	0.62	0.40	0.67	**0.77**	0.43	0.66	0.40	0.29
LFR, sizes 50–250		0.42$^-$	0.43	0.44	0.16	0.46	0.16	**0.87**	0.85	0.77	0.75	0.38	0.79	0.62	0.33	0.55	0.14	0.08
LFR, C_{LFR}=0.3		0.39$^-$	0.40	0.39	0.24	0.31	0.42	**0.62**	0.61	0.58	0.52	0.48	0.48	0.59	0.27	0.36	0.23	0.22
LFR, C_{LFR}=0.5		0.30$^-$	0.30	0.29	0.24	0.25	0.28	**0.38**	**0.37**	0.35	0.26	0.35	0.29	0.37	0.22	0.27	0.22	0.23

TABLE S4. As Table S1, with each method's parameter frozen on three calibration graphs per noise level of the same family, all noise levels pooled (frozen protocol). 95% bootstrap half-widths of these means over the ten test seeds have median 0.005 and maximum 0.069. Dashes: the level of the stochastic block model's hierarchy frozen on the calibration graphs does not exist on some test graphs of that family, whose fitted hierarchy has fewer levels; on these rows the best other method is taken over the remaining methods, which cannot change a mark, since the stochastic block model's oracle Ω there is far below that of the best other method (Table S3). Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.58$^-$	0.58	0.51	0.52	0.58	0.51	**0.66**	**0.66**	0.42	0.39	0.14	0.41	0.47	0.53	0.53	0.48	0.52
	0.05	0.51$^-$	0.51	0.49	0.11	0.52	0.49	**0.54**	0.53	0.31	0.24	0.12	–	0.44	0.43	0.42	0.10	0.13
	0.10	**0.44**$^=$	**0.44**	0.44	0.08	0.43	0.44	0.40	0.39	0.27	0.18	0.11	0.25	0.40	0.40	0.41	0.07	0.09
uniform, k=16	0.00	0.74$^-$	0.74	0.71	0.67	0.70	0.64	**0.85**	**0.84**	0.37	0.65	0.14	0.53	0.42	0.64	0.67	0.63	0.58
	0.05	0.56$^-$	0.58	0.57	0.13	0.36	0.52	**0.76**	0.75	0.33	0.47	0.11	0.41	0.39	0.48	0.63	0.13	0.16
	0.10	0.31$^-$	0.32	0.31	0.10	0.24	0.52	**0.62**	0.62	0.28	0.33	0.09	0.34	0.36	0.42	0.57	0.10	0.12
equal k, unequal s, k=8	0.00	0.52$^-$	0.53	0.46	0.48	0.53	0.46	**0.56**	**0.56**	0.29	0.33	0.10	0.39	0.35	0.48	0.48	0.44	0.48
	0.05	**0.46**$^=$	0.45	0.43	0.09	**0.46**	0.43	0.38	0.38	0.22	0.19	0.09	0.28	0.32	0.38	0.34	0.08	0.11
	0.10	0.36$^=$	**0.36**	0.35	0.07	0.35	0.36	0.28	0.28	0.19	0.15	0.08	0.22	0.29	0.35	0.33	0.05	0.08
equal k, unequal s, k=16	0.00	0.76$^-$	0.76	0.73	0.69	0.73	0.68	**0.86**	0.85	0.55	0.69	0.17	0.46	0.51	0.66	0.69	0.64	0.59
	0.05	0.60$^-$	0.61	0.60	0.14	0.40	0.54	**0.77**	0.76	0.50	0.56	0.14	0.44	0.48	0.53	0.66	0.13	0.17
	0.10	0.33$^-$	0.36	0.37	0.10	0.28	0.54	**0.64**	**0.64**	0.43	0.44	0.12	0.38	0.44	0.45	0.61	0.10	0.13
equal s, unequal k	0.00	0.74$^-$	0.74	0.74	0.65	0.73	0.59	**0.81**	0.77	0.51	0.60	0.16	0.53	0.37	0.53	0.64	0.61	0.57
	0.05	0.52$^-$	0.53	0.56	0.17	0.32	0.59	**0.70**	**0.70**	0.44	0.52	0.13	0.44	0.33	0.40	0.57	0.14	0.17
	0.10	0.30$^-$	0.30	0.29	0.12	0.22	0.53	0.62	**0.63**	0.38	0.46	0.11	0.36	0.29	0.34	0.46	0.12	0.15
two worlds	0.00	0.77$^-$	0.77	0.79	0.66	0.77	0.68	0.66	**0.82**	0.33	0.52	0.13	0.31	0.28	0.49	0.65	0.64	0.60
	0.05	0.51$^-$	0.55	0.57	0.18	0.30	0.64	0.54	**0.76**	0.28	0.41	0.10	0.26	0.25	0.38	0.55	0.16	0.20
	0.10	0.33$^-$	0.34	0.32	0.15	0.24	0.52	0.40	**0.71**	0.26	0.32	0.08	0.21	0.22	0.33	0.45	0.14	0.17
three big, many small	0.00	0.80$^+$	**0.81**	0.80	0.70	0.77	0.70	0.35	0.67	0.19	0.27	0.07	–	0.16	0.45	0.64	0.64	0.59
	0.05	0.45$^-$	0.49	0.51	0.21	0.29	**0.61**	0.29	0.56	0.15	0.22	0.06	0.16	0.14	0.35	0.45	0.18	0.19
	0.10	0.29$^-$	0.30	0.29	0.16	0.24	**0.52**	0.23	0.41	0.14	0.17	0.05	0.13	0.12	0.31	0.41	0.13	0.16
gradient	0.00	**0.60**$^=$	0.59	0.57	0.53	**0.60**	0.57	0.23	0.32	0.08	0.14	0.03	0.38	0.11	0.50	0.43	0.52	0.51
	0.05	0.40$^=$	0.40	0.38	0.12	0.24	0.33	0.14	0.20	0.07	0.10	0.03	0.09	0.09	0.40	**0.42**	0.08	0.10
	0.10	0.20$^-$	0.23	0.22	0.10	0.17	0.32	0.09	0.14	0.06	0.07	0.02	0.01	0.08	**0.34**	0.32	0.07	0.08
uniform k=8, α=1	0.05	0.52$^-$	0.51	0.48	0.11	0.52	0.48	**0.54**	0.54	0.31	0.24	0.12	0.31	0.44	0.43	0.42	0.10	0.13
uniform k=8, α=2	0.05	0.52$^-$	0.52	0.48	0.12	0.52	0.48	**0.54**	0.54	0.31	0.22	0.12	0.33	0.43	0.43	0.42	0.10	0.14
two worlds, α=1	0.05	0.51$^-$	0.54	0.56	0.16	0.30	0.58	0.59	**0.79**	0.29	0.62	0.11	0.28	0.27	0.34	0.44	0.15	0.19
two worlds, α=2	0.05	0.55$^-$	0.59	0.58	0.20	0.33	0.52	0.64	**0.75**	0.32	0.62	0.10	–	0.29	0.31	0.42	0.17	0.19
LFR, o_n=0		0.82$^-$	0.83	0.83	0.62	0.80	0.66	**1.00**	**1.00**	**1.00**	0.98	0.61	0.99	0.91	0.47	0.77	0.53	0.40
LFR, o_n=0.1		0.64$^-$	0.68	0.67	0.14	0.48	0.60	**0.88**	**0.88**	**0.88**	0.81	0.49	0.87	0.82	0.44	0.72	0.17	0.19
LFR, o_n=0.3		0.29$^-$	0.37	0.34	0.05	0.23	0.48	**0.69**	**0.69**	0.64	0.49	0.32	0.57	0.67	0.40	0.64	0.07	0.05
LFR, $\langle k \rangle$=10		0.42$^-$	0.45	0.36	0.11	0.47	0.36	**0.79**	**0.79**	0.71	0.61	0.36	0.63	0.53	0.26	0.31	0.12	0.10
LFR, o_m=4		0.55$^-$	0.57	0.60	0.31	0.49	0.52	0.71	0.71	0.71	0.62	0.40	0.67	**0.73**	0.43	0.66	0.32	0.28
LFR, sizes 50–250		0.32$^-$	0.32	0.37	0.09	0.35	0.16	**0.87**	0.84	0.77	0.75	0.38	0.79	0.60	0.33	0.55	0.08	0.04
LFR, C_{LFR}=0.3		0.30$^-$	0.30	0.29	0.21	0.24	0.32	**0.61**	**0.60**	0.58	0.52	0.48	0.46	0.58	0.25	0.35	0.19	0.21
LFR, C_{LFR}=0.5		0.23$^-$	0.23	0.23	0.22	0.22	0.27	0.36	0.35	0.34	0.26	0.35	0.25	**0.37**	0.22	0.27	0.21	0.20

TABLE S5. As Table S1, with each method at its own default or selection rule: Leiden at $\gamma = 1$, L-CPM at γ equal to the edge density, link communities at maximum partition density, Infomap at Markov time 1, SLPA at $r = 0.1$, SBM at the finest level of its hierarchy, and the label-free choice for the methods without one (native protocol). 95% bootstrap half-widths of these means over the ten test seeds have median 0.006 and maximum 0.051. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	**0.61**$^{=}$	0.61	0.51	0.51	0.25	0.51	**0.61**	0.61	0.36	0.39	0.14	0.41	0.42	0.53	0.52	0.50	0.53
	0.05	**0.55**$^{+}$	0.54	0.49	0.29	0.24	0.49	0.45	0.44	0.31	0.24	0.12	0.30	0.39	0.44	0.39	0.26	0.30
	0.10	**0.50**$^{+}$	**0.50**	0.46	0.22	0.23	0.44	0.32	0.32	0.27	0.18	0.11	0.24	0.36	0.40	0.40	0.21	0.21
uniform, k=16	0.00	0.78^{-}	0.78	0.75	0.69	0.31	0.67	**0.84**	0.84	0.37	0.65	0.14	0.22	0.40	0.62	0.67	0.65	0.59
	0.05	0.72^{-}	0.71	0.69	0.39	0.26	0.56	**0.75**	0.75	0.33	0.47	0.11	0.20	0.37	0.50	0.63	0.33	0.35
	0.10	**0.62**$^{=}$	**0.62**	0.60	0.29	0.23	0.54	**0.62**	0.61	0.28	0.33	0.09	0.17	0.33	0.45	0.57	0.24	0.25
equal k, unequal s, k=8	0.00	**0.57**$^{+}$	0.57	0.46	0.49	0.18	0.46	0.56	0.56	0.25	0.33	0.10	0.28	0.30	0.48	0.49	0.47	0.49
	0.05	**0.52**$^{+}$	0.51	0.45	0.25	0.18	0.43	0.35	0.35	0.22	0.19	0.09	0.22	0.28	0.38	0.35	0.22	0.23
	0.10	**0.45**$^{+}$	**0.45**	0.40	0.17	0.16	0.37	0.25	0.24	0.19	0.15	0.08	0.17	0.26	0.33	0.33	0.15	0.16
equal k, unequal s, k=16	0.00	0.80^{-}	0.79	0.77	0.70	0.36	0.71	**0.84**	**0.84**	0.55	0.69	0.17	0.35	0.49	0.64	0.69	0.66	0.60
	0.05	**0.73**$^{+}$	**0.73**	0.71	0.42	0.33	0.59	0.72	0.72	0.50	0.56	0.14	0.29	0.46	0.51	0.66	0.37	0.37
	0.10	**0.65**$^{+}$	**0.65**	0.62	0.31	0.30	0.57	0.59	0.59	0.43	0.44	0.12	0.24	0.42	0.45	0.61	0.27	0.29
equal s, unequal k	0.00	**0.78**$^{=}$	**0.78**	0.76	0.71	0.47	0.60	**0.78**	0.75	0.51	0.60	0.16	0.24	0.37	0.53	0.64	0.66	0.61
	0.05	**0.70**$^{+}$	**0.70**	0.69	0.47	0.41	0.59	0.67	0.67	0.44	0.52	0.13	0.21	0.33	0.38	0.57	0.40	0.41
	0.10	**0.61**$^{+}$	**0.61**	0.59	0.39	0.38	0.56	0.58	0.58	0.38	0.46	0.11	0.18	0.29	0.29	0.46	0.32	0.34
two worlds	0.00	**0.83**$^{+}$	**0.82**	0.82	0.71	0.35	0.70	0.60	0.82	0.33	0.52	0.13	0.19	0.28	0.48	0.65	0.72	0.64
	0.05	**0.76**$^{=}$	**0.76**	**0.76**	0.58	0.33	0.66	0.44	**0.76**	0.28	0.41	0.10	0.13	0.25	0.35	0.55	0.51	0.49
	0.10	**0.71**$^{=}$	**0.71**	0.70	0.49	0.28	0.65	0.38	0.70	0.26	0.32	0.08	0.10	0.21	0.28	0.43	0.42	0.43
three big, many small	0.00	**0.83**$^{+}$	**0.83**	0.82	0.76	0.25	0.72	0.33	0.51	0.19	0.27	0.07	0.07	0.16	0.47	0.64	0.74	0.66
	0.05	**0.78**$^{+}$	**0.78**	0.74	0.48	0.21	0.68	0.28	0.43	0.15	0.22	0.06	0.05	0.14	0.35	0.51	0.42	0.44
	0.10	**0.67**$^{+}$	0.66	0.62	0.36	0.18	0.61	0.23	0.37	0.14	0.17	0.05	0.03	0.11	0.29	0.43	0.33	0.40
gradient	0.00	0.42^{-}	0.40	0.39	0.36	0.11	**0.56**	0.21	0.23	0.08	0.14	0.03	0.06	0.09	0.50	0.48	0.41	0.38
	0.05	0.32^{-}	0.31	0.27	0.20	0.08	0.40	0.14	0.17	0.07	0.10	0.03	0.04	0.08	0.34	**0.42**	0.18	0.16
	0.10	0.23^{-}	0.24	0.22	0.13	0.07	**0.35**	0.09	0.12	0.06	0.07	0.02	0.03	0.07	0.27	0.33	0.12	0.13
uniform k=8, α=1	0.05	**0.56**$^{+}$	0.55	0.49	0.31	0.24	0.48	0.45	0.44	0.31	0.24	0.12	0.31	0.39	0.43	0.37	0.28	0.32
uniform k=8, α=2	0.05	**0.56**$^{+}$	**0.56**	0.49	0.33	0.23	0.48	0.45	0.44	0.31	0.22	0.12	0.32	0.39	0.43	0.39	0.30	0.32
two worlds, α=1	0.05	**0.77**$^{+}$	**0.77**	0.75	0.54	0.28	0.65	0.47	0.56	0.29	0.62	0.11	0.19	0.27	0.26	0.44	0.46	0.47
two worlds, α=2	0.05	**0.79**$^{+}$	**0.79**	0.75	0.51	0.28	0.64	0.49	0.52	0.32	0.62	0.10	0.22	0.27	0.23	0.41	0.50	0.50
LFR, o_n=0		0.84^{-}	0.86	0.84	0.56	0.82	0.66	0.98	0.97	**1.00**	0.98	0.61	0.99	0.89	0.46	0.77	0.43	0.39
LFR, o_n=0.1		0.69^{-}	0.72	0.71	0.27	0.56	0.60	0.78	0.78	**0.88**	0.81	0.49	0.87	0.82	0.41	0.72	0.29	0.23
LFR, o_n=0.3		0.45^{-}	0.50	0.46	0.11	0.34	0.48	0.54	0.53	0.64	0.49	0.32	0.57	**0.67**	0.40	0.64	0.13	0.10
LFR, $\langle k \rangle$=10		0.43^{-}	0.45	0.35	0.09	0.33	0.36	0.66	0.63	**0.71**	0.61	0.36	0.63	0.53	0.26	0.25	0.12	0.12
LFR, o_m=4		0.61^{-}	0.63	0.62	0.40	0.52	0.52	0.62	0.63	**0.71**	0.62	0.40	0.67	**0.71**	0.43	0.66	0.36	0.28
LFR, sizes 50–250		0.42^{-}	0.41	0.42	0.14	0.38	0.16	**0.84**	0.80	0.77	0.75	0.38	0.79	0.60	0.18	0.55	0.13	0.07
LFR, C_{LFR}=0.3		0.33^{-}	0.33	0.32	0.19	0.28	0.39	0.50	0.50	**0.58**	0.52	0.48	0.46	**0.58**	0.18	0.30	0.20	0.20
LFR, C_{LFR}=0.5		0.24^{-}	0.24	0.24	0.19	0.22	0.24	0.29	0.29	0.34	0.26	0.35	0.25	**0.36**	0.17	0.21	0.18	0.20

TABLE S6. ONMI (McDaid *et al.*, max normalisation) of the oracle covers of Table S3; superscripts on HC as in Table S1, for ONMI. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.52$^-$	0.51	0.41	0.47	0.49	0.41	**0.61**	**0.61**	0.36	0.38	0.00	0.34	0.15	0.44	0.46	0.45	0.47
	0.05	**0.47**$^+$	0.46	0.39	0.20	0.42	0.39	0.46	0.45	0.18	0.22	0.00	0.23	0.12	0.28	0.28	0.16	0.19
	0.10	**0.41**$^+$	0.40	0.37	0.12	0.33	0.35	0.27	0.27	0.11	0.11	0.00	0.14	0.10	0.23	0.27	0.09	0.10
uniform, k=16	0.00	0.71$^-$	0.71	0.67	0.65	0.51	0.59	**0.83**	0.82	0.31	0.62	0.01	0.41	0.09	0.53	0.56	0.59	0.54
	0.05	0.66$^-$	0.65	0.62	0.29	0.32	0.46	**0.71**	**0.70**	0.16	0.42	0.01	0.25	0.09	0.29	0.52	0.22	0.26
	0.10	**0.56**$^=$	**0.55**	0.52	0.19	0.18	0.45	**0.55**	0.54	0.11	0.26	0.01	0.14	0.08	0.22	0.46	0.13	0.17
equal k, unequal s, k=8	0.00	0.49$^-$	0.48	0.38	0.44	0.46	0.38	**0.55**	**0.54**	0.24	0.28	0.00	0.25	0.08	0.41	0.42	0.41	0.43
	0.05	**0.44**$^+$	0.43	0.36	0.15	0.38	0.35	0.31	0.31	0.11	0.13	0.00	0.14	0.06	0.24	0.23	0.11	0.14
	0.10	**0.36**$^+$	0.36	0.32	0.08	0.30	0.30	0.17	0.17	0.06	0.06	0.00	0.08	0.05	0.20	0.21	0.05	0.08
equal k, unequal s, k=16	0.00	0.74$^-$	0.73	0.69	0.66	0.53	0.62	**0.85**	**0.85**	0.53	0.71	0.07	0.42	0.26	0.57	0.60	0.61	0.55
	0.05	0.69$^-$	0.68	0.65	0.36	0.37	0.50	**0.74**	0.73	0.42	0.56	0.06	0.34	0.26	0.39	0.57	0.29	0.31
	0.10	**0.61**$^=$	**0.61**	0.57	0.24	0.23	0.48	**0.61**	0.60	0.36	0.44	0.05	0.26	0.24	0.30	0.52	0.18	0.21
equal s, unequal k	0.00	0.67$^-$	0.67	0.61	0.62	0.38	0.42	**0.76**	0.70	0.38	0.56	0.05	0.49	0.14	0.40	0.46	0.58	0.52
	0.05	0.57$^-$	0.57	0.53	0.31	0.21	0.40	**0.64**	0.58	0.27	0.47	0.04	0.37	0.13	0.21	0.39	0.25	0.28
	0.10	0.45$^-$	0.44	0.42	0.24	0.12	0.36	**0.52**	0.51	0.23	0.40	0.04	0.27	0.11	0.15	0.27	0.19	0.22
two worlds	0.00	0.62$^-$	0.60	0.54	0.56	0.34	0.32	0.54	**0.70**	0.26	0.38	0.01	0.24	0.09	0.34	0.38	0.53	0.48
	0.05	0.42$^-$	0.43	0.40	0.22	0.15	0.28	0.35	**0.57**	0.14	0.25	0.01	0.17	0.06	0.15	0.28	0.19	0.21
	0.10	0.28$^-$	0.28	0.27	0.17	0.08	0.24	0.24	**0.43**	0.09	0.18	0.01	0.13	0.04	0.11	0.16	0.14	0.16
three big, many small	0.00	0.62$^-$	0.61	0.55	0.57	0.30	0.33	0.32	**0.69**	0.15	0.19	0.00	0.19	0.06	0.29	0.36	0.54	0.48
	0.05	0.40$^-$	0.40	0.38	0.19	0.11	0.29	0.18	**0.52**	0.07	0.14	0.00	0.11	0.04	0.13	0.21	0.17	0.20
	0.10	0.26$^=$	**0.26**	0.25	0.15	0.06	0.24	0.11	**0.27**	0.05	0.10	0.00	0.06	0.02	0.09	0.12	0.12	0.15
gradient	0.00	**0.51**$^+$	0.50	0.43	0.46	0.42	0.40	0.29	0.34	0.08	0.05	0.00	0.12	0.01	0.37	0.32	0.45	0.44
	0.05	**0.38**$^+$	**0.37**	0.33	0.12	0.18	0.20	0.09	0.11	0.02	0.03	0.00	0.04	0.01	0.16	0.25	0.08	0.11
	0.10	**0.23**$^+$	**0.23**	0.20	0.08	0.08	0.19	0.03	0.05	0.01	0.02	0.00	0.03	0.01	0.10	0.17	0.05	0.07
uniform k=8, α=1	0.05	**0.47**$^+$	0.46	0.39	0.22	0.42	0.39	0.46	0.46	0.18	0.21	0.00	0.24	0.13	0.28	0.28	0.16	0.20
uniform k=8, α=2	0.05	**0.48**$^+$	0.47	0.39	0.25	0.41	0.38	0.46	0.46	0.18	0.20	0.00	0.24	0.12	0.28	0.28	0.19	0.23
two worlds, α=1	0.05	0.51$^-$	0.51	0.44	0.31	0.17	0.26	0.39	**0.58**	0.18	0.23	0.01	0.17	0.08	0.13	0.22	0.24	0.28
two worlds, α=2	0.05	**0.60**$^+$	**0.59**	0.47	0.41	0.18	0.25	0.44	0.56	0.21	0.22	0.01	0.17	0.09	0.11	0.20	0.32	0.35
LFR, o_n=0		0.80$^-$	0.80	0.77	0.60	0.38	0.54	**1.00**	**1.00**	**1.00**	0.98	0.65	0.99	0.92	0.37	0.69	0.52	0.37
LFR, o_n=0.1		0.70$^-$	0.71	0.68	0.28	0.24	0.51	**0.87**	**0.87**	**0.87**	0.79	0.49	0.84	0.85	0.28	0.64	0.29	0.22
LFR, o_n=0.3		0.49$^-$	0.51	0.49	0.12	0.12	0.45	0.69	0.69	0.66	0.55	0.28	0.58	**0.70**	0.21	0.53	0.14	0.10
LFR, $\langle k \rangle$=10		0.37$^-$	0.37	0.26	0.08	0.29	0.25	**0.77**	**0.77**	0.69	0.57	0.36	0.62	0.37	0.22	0.23	0.11	0.09
LFR, o_m=4		0.59$^-$	0.60	0.59	0.41	0.20	0.50	0.69	0.69	0.69	0.60	0.37	0.62	**0.73**	0.35	0.59	0.39	0.30
LFR, sizes 50–250		0.34$^-$	0.35	0.30	0.11	0.12	0.10	**0.84**	0.83	0.74	0.72	0.37	0.77	0.63	0.15	0.42	0.10	0.06
LFR, C_{LFR}=0.3		0.31$^-$	0.32	0.31	0.17	0.08	0.31	**0.57**	0.56	0.53	0.48	0.45	0.40	0.51	0.13	0.20	0.17	0.16
LFR, C_{LFR}=0.5		0.19$^-$	0.19	0.18	0.13	0.04	0.13	**0.30**	0.28	0.25	0.18	0.26	0.16	0.22	0.08	0.08	0.12	0.13

TABLE S7. Per-community recovery over all true communities (mean over true communities of the Jaccard similarity with the best-matching detected community) of the oracle covers; superscripts as in Table S1, for this score. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.63$^-$	0.62	0.52	0.57	0.62	0.52	**0.69**	**0.69**	0.48	0.49	0.15	0.39	0.39	0.56	0.55	0.56	0.56
	0.05	**0.58**$^+$	0.57	0.51	0.36	0.57	0.50	0.56	0.56	0.35	0.36	0.14	0.29	0.37	0.48	0.43	0.32	0.34
	0.10	**0.53**$^+$	0.52	0.48	0.29	0.52	0.46	0.42	0.42	0.29	0.28	0.13	0.24	0.35	0.44	0.41	0.26	0.26
uniform, k=16	0.00	0.79$^-$	0.79	0.76	0.72	0.79	0.69	**0.86**	**0.86**	0.44	0.69	0.15	0.48	0.34	0.71	0.67	0.67	0.61
	0.05	0.73$^-$	0.73	0.70	0.44	0.68	0.57	**0.78**	**0.77**	0.31	0.52	0.14	0.37	0.33	0.61	0.63	0.37	0.39
	0.10	**0.65**$^=$	0.64	0.62	0.35	0.56	0.55	**0.65**	0.64	0.27	0.38	0.12	0.30	0.31	0.54	0.58	0.29	0.31
equal k, unequal s, k=8	0.00	0.60$^-$	0.60	0.49	0.54	0.60	0.49	**0.64**	**0.64**	0.41	0.43	0.14	0.33	0.33	0.54	0.53	0.53	0.53
	0.05	**0.56**$^+$	0.55	0.48	0.32	0.54	0.47	0.46	0.46	0.30	0.30	0.13	0.26	0.32	0.45	0.39	0.29	0.30
	0.10	**0.49**$^=$	0.48	0.44	0.25	**0.49**	0.42	0.35	0.35	0.25	0.24	0.12	0.21	0.30	0.42	0.37	0.23	0.23
equal k, unequal s, k=16	0.00	0.81$^-$	0.81	0.79	0.74	0.81	0.74	**0.89**	**0.88**	0.67	0.78	0.20	0.50	0.51	0.75	0.72	0.70	0.64
	0.05	0.77$^-$	0.77	0.74	0.51	0.72	0.63	**0.81**	0.80	0.59	0.67	0.18	0.44	0.49	0.69	0.68	0.45	0.45
	0.10	**0.70**$^=$	**0.70**	0.67	0.41	0.63	0.60	**0.70**	0.68	0.53	0.57	0.15	0.38	0.47	0.64	0.64	0.36	0.37
equal s, unequal k	0.00	0.77$^-$	0.77	0.73	0.72	0.75	0.57	**0.80**	0.78	0.50	0.65	0.17	0.53	0.33	0.60	0.61	0.68	0.62
	0.05	0.68$^-$	0.69	0.66	0.47	0.61	0.55	**0.70**	0.68	0.40	0.56	0.15	0.45	0.30	0.48	0.55	0.41	0.42
	0.10	0.58$^-$	0.58	0.56	0.40	0.52	0.51	**0.61**	**0.61**	0.35	0.51	0.14	0.37	0.28	0.42	0.47	0.33	0.35
two worlds	0.00	0.67$^-$	0.66	0.59	0.61	0.63	0.37	0.52	**0.72**	0.39	0.42	0.14	0.25	0.29	0.49	0.47	0.59	0.54
	0.05	0.48$^-$	0.48	0.45	0.29	0.40	0.31	0.34	**0.60**	0.29	0.33	0.12	0.20	0.26	0.35	0.39	0.27	0.29
	0.10	0.33$^-$	0.33	0.31	0.24	0.31	0.26	0.24	**0.46**	0.25	0.27	0.10	0.18	0.23	0.32	0.30	0.22	0.24
three big, many small	0.00	0.65$^-$	0.64	0.58	0.61	0.61	0.34	0.28	**0.71**	0.32	0.31	0.13	0.15	0.28	0.45	0.43	0.57	0.53
	0.05	0.41$^-$	0.42	0.40	0.25	0.35	0.29	0.17	**0.56**	0.26	0.25	0.12	0.17	0.24	0.34	0.33	0.24	0.27
	0.10	0.26$^-$	0.26	0.25	0.22	0.27	0.23	0.16	**0.38**	0.23	0.21	0.11	0.14	0.21	0.30	0.25	0.20	0.22
gradient	0.00	**0.60**$^+$	0.59	0.51	0.55	0.58	0.49	0.36	0.48	0.26	0.26	0.13	0.23	0.23	0.49	0.43	0.52	0.51
	0.05	**0.48**$^+$	**0.48**	0.43	0.27	0.41	0.32	0.24	0.28	0.19	0.21	0.12	0.19	0.22	0.39	0.37	0.22	0.24
	0.10	**0.36**$^+$	**0.35**	0.32	0.22	0.33	0.30	0.18	0.21	0.17	0.18	0.12	0.17	0.20	0.33	0.31	0.18	0.18
uniform k=8, α=1	0.05	**0.58**$^+$	0.57	0.50	0.37	0.57	0.50	0.57	0.56	0.35	0.35	0.14	0.31	0.37	0.48	0.42	0.33	0.35
uniform k=8, α=2	0.05	**0.59**$^+$	0.58	0.50	0.40	0.57	0.49	0.57	0.56	0.34	0.34	0.14	0.31	0.37	0.48	0.43	0.35	0.38
two worlds, α=1	0.05	0.57$^-$	0.57	0.49	0.40	0.49	0.29	0.39	**0.61**	0.33	0.30	0.12	0.22	0.28	0.33	0.34	0.36	0.38
two worlds, α=2	0.05	**0.66**$^+$	**0.65**	0.52	0.48	0.58	0.29	0.44	0.59	0.37	0.27	0.13	0.22	0.28	0.32	0.32	0.44	0.46
LFR, o_n=0		0.90$^-$	0.91	0.91	0.76	0.90	0.80	**1.00**	**1.00**	**1.00**	0.99	0.66	0.99	0.93	0.75	0.82	0.72	0.60
LFR, o_n=0.1		0.81$^-$	0.83	0.81	0.49	0.74	0.75	**0.92**	**0.92**	**0.92**	0.85	0.53	0.90	0.88	0.71	0.79	0.50	0.43
LFR, o_n=0.3		0.64$^-$	0.66	0.65	0.30	0.56	0.65	**0.78**	**0.78**	0.76	0.67	0.34	0.67	0.76	0.66	0.73	0.32	0.27
LFR, $\langle k \rangle$=10		0.58$^-$	0.59	0.52	0.26	0.62	0.52	**0.83**	0.83	0.78	0.66	0.41	0.66	0.61	0.50	0.48	0.30	0.27
LFR, o_m=4		0.70$^-$	0.72	0.71	0.55	0.68	0.67	0.78	0.77	0.78	0.67	0.43	0.70	**0.80**	0.70	0.74	0.53	0.46
LFR, sizes 50–250		0.56$^-$	0.56	0.59	0.30	0.58	0.35	**0.91**	0.90	0.81	0.81	0.43	0.84	0.70	0.55	0.64	0.31	0.23
LFR, C_{LFR}=0.3		0.48$^-$	0.49	0.48	0.37	0.44	0.53	**0.66**	0.65	0.65	0.59	0.51	0.55	0.64	0.56	0.50	0.37	0.37
LFR, C_{LFR}=0.5		0.39$^-$	0.39	0.38	0.34	0.39	0.37	0.46	0.45	0.43	0.36	0.43	0.35	0.45	**0.50**	0.42	0.33	0.34

TABLE S8. ONMI (McDaid *et al.*, max normalisation) of the label-free covers of Table S1; superscripts on HC as in Table S1, for ONMI. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.52$^-$	0.51	0.41	0.46	0.49	0.41	**0.57**	**0.57**	0.37	0.38	0.00	0.31	0.07	0.44	0.46	0.45	0.47
	0.05	**0.47**$^+$	0.46	0.39	0.20	0.43	0.39	0.36	0.35	0.17	0.22	0.00	0.16	0.06	0.28	0.29	0.17	0.19
	0.10	**0.41**$^+$	0.41	0.36	0.12	0.36	0.35	0.20	0.19	0.10	0.11	0.00	0.10	0.04	0.22	0.27	0.09	0.11
uniform, k=16	0.00	0.72$^-$	0.71	0.67	0.65	0.52	0.60	**0.82**	0.82	0.31	0.62	0.01	0.41	0.05	0.53	0.56	0.59	0.54
	0.05	0.66$^-$	0.65	0.62	0.30	0.37	0.46	**0.70**	**0.69**	0.16	0.42	0.01	0.24	0.05	0.33	0.53	0.23	0.26
	0.10	**0.56**$^+$	**0.55**	0.52	0.19	0.24	0.45	0.53	0.52	0.11	0.26	0.01	0.14	0.05	0.25	0.47	0.14	0.17
equal k, unequal s, k=8	0.00	0.49$^-$	0.48	0.38	0.44	0.46	0.38	**0.52**	0.51	0.24	0.28	0.00	0.25	0.03	0.40	0.42	0.42	0.44
	0.05	**0.44**$^+$	0.43	0.36	0.16	0.40	0.35	0.27	0.27	0.11	0.13	0.00	0.13	0.03	0.22	0.23	0.12	0.14
	0.10	**0.37**$^+$	0.35	0.32	0.09	0.33	0.30	0.13	0.13	0.05	0.06	0.00	0.06	0.02	0.16	0.21	0.06	0.08
equal k, unequal s, k=16	0.00	0.74$^-$	0.73	0.69	0.67	0.54	0.63	**0.82**	**0.82**	0.53	0.71	0.07	0.40	0.23	0.56	0.60	0.61	0.55
	0.05	**0.68**$^+$	**0.68**	0.64	0.36	0.40	0.50	0.67	0.67	0.41	0.56	0.06	0.31	0.23	0.43	0.57	0.31	0.30
	0.10	**0.60**$^+$	**0.60**	0.57	0.23	0.29	0.48	0.51	0.50	0.35	0.44	0.05	0.22	0.21	0.37	0.52	0.19	0.21
equal s, unequal k	0.00	0.67$^-$	0.67	0.61	0.63	0.36	0.40	**0.76**	0.71	0.39	0.56	0.05	0.49	0.14	0.46	0.46	0.59	0.52
	0.05	0.55$^-$	0.54	0.51	0.31	0.23	0.40	**0.61**	0.58	0.28	0.47	0.04	0.36	0.13	0.31	0.40	0.26	0.28
	0.10	0.44$^-$	0.45	0.42	0.24	0.13	0.37	0.49	**0.50**	0.23	0.40	0.04	0.27	0.12	0.25	0.35	0.19	0.22
two worlds	0.00	0.62$^-$	0.61	0.54	0.56	0.29	0.31	0.48	**0.67**	0.26	0.38	0.01	0.23	0.05	0.39	0.39	0.53	0.48
	0.05	**0.45**$^=$	**0.45**	0.41	0.22	0.17	0.30	0.29	0.45	0.14	0.25	0.01	0.17	0.04	0.26	0.34	0.19	0.21
	0.10	**0.28**$^+$	**0.29**	0.27	0.17	0.09	0.25	0.19	0.27	0.10	0.18	0.01	0.13	0.03	0.20	0.26	0.14	0.16
three big, many small	0.00	**0.63**$^+$	0.62	0.55	0.57	0.25	0.32	0.32	0.37	0.14	0.19	0.00	0.15	0.03	0.33	0.38	0.54	0.48
	0.05	0.39$^+$	**0.40**	0.37	0.17	0.13	0.28	0.19	0.27	0.07	0.14	0.00	0.09	0.02	0.13	0.26	0.15	0.17
	0.10	**0.25**$^+$	**0.24**	0.23	0.12	0.06	0.22	0.11	0.18	0.05	0.10	0.00	0.06	0.01	0.09	0.15	0.11	0.14
gradient	0.00	0.27$^-$	0.24	0.25	0.23	**0.40**	0.38	0.20	0.16	0.05	0.05	0.00	0.11	0.01	0.35	0.29	0.31	0.26
	0.05	0.15$^-$	0.14	0.11	0.07	0.19	0.18	0.08	0.09	0.02	0.03	0.00	0.03	0.01	0.09	**0.23**	0.07	0.08
	0.10	0.07$^-$	0.07	0.06	0.03	0.09	**0.16**	0.03	0.04	0.01	0.02	0.00	0.02	0.01	0.05	0.15	0.04	0.05
uniform k=8, α=1	0.05	**0.47**$^+$	0.46	0.39	0.22	0.43	0.39	0.36	0.36	0.18	0.21	0.00	0.19	0.05	0.28	0.28	0.18	0.21
uniform k=8, α=2	0.05	**0.48**$^+$	0.47	0.39	0.25	0.43	0.38	0.36	0.36	0.17	0.20	0.00	0.18	0.05	0.28	0.29	0.21	0.23
two worlds, α=1	0.05	**0.52**$^+$	**0.51**	0.43	0.31	0.19	0.28	0.33	0.44	0.16	0.23	0.01	0.16	0.05	0.24	0.30	0.24	0.28
two worlds, α=2	0.05	**0.59**$^+$	**0.59**	0.46	0.43	0.19	0.28	0.37	0.44	0.20	0.22	0.01	0.17	0.06	0.22	0.27	0.33	0.36
LFR, o_n=0		0.80$^-$	0.80	0.77	0.57	0.38	0.54	0.97	0.97	**1.00**	0.98	0.65	0.99	0.92	0.47	0.70	0.49	0.36
LFR, o_n=0.1		0.66$^-$	0.69	0.67	0.26	0.24	0.51	0.68	0.68	**0.86**	0.79	0.49	0.84	0.79	0.41	0.66	0.28	0.22
LFR, o_n=0.3		0.44$^-$	0.48	0.44	0.10	0.12	0.45	0.35	0.35	**0.64**	0.55	0.28	0.40	0.59	0.38	0.57	0.13	0.10
LFR, $\langle k \rangle$=10		0.36$^-$	0.37	0.26	0.07	0.30	0.25	0.58	0.60	**0.68**	0.57	0.36	0.62	0.40	0.23	0.19	0.10	0.09
LFR, o_m=4		0.58$^-$	0.60	0.58	0.38	0.19	0.50	0.47	0.47	**0.68**	0.60	0.37	0.62	0.64	0.42	0.61	0.37	0.31
LFR, sizes 50–250		0.35$^-$	0.34	0.30	0.10	0.13	0.10	**0.84**	0.82	0.74	0.72	0.37	0.77	0.59	0.17	0.42	0.10	0.06
LFR, C_{LFR}=0.3		0.28$^-$	0.28	0.28	0.15	0.08	0.29	0.35	0.35	0.48	0.48	0.45	0.23	**0.50**	0.19	0.23	0.16	0.16
LFR, C_{LFR}=0.5		0.16$^-$	0.16	0.15	0.11	0.04	0.11	0.12	0.12	0.17	0.18	**0.26**	0.10	0.24	0.10	0.10	0.10	0.12

TABLE S9. Per-community recovery over all true communities (mean over true communities of the Jaccard similarity with the best-matching detected community) of the label-free covers; superscripts as in Table S1, for this score. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	**0.62**$^{=}$	**0.62**	0.52	0.55	0.62	0.52	**0.62**	0.62	0.49	0.49	0.15	0.35	0.31	0.56	0.55	0.54	0.55
	0.05	**0.57**$^{+}$	0.56	0.50	0.34	0.55	0.50	0.43	0.42	0.33	0.36	0.14	0.23	0.30	0.48	0.41	0.31	0.33
	0.10	**0.52**$^{+}$	**0.51**	0.47	0.27	0.48	0.45	0.30	0.30	0.28	0.28	0.13	0.20	0.28	0.43	0.41	0.26	0.26
uniform, k=16	0.00	0.79^{-}	0.79	0.75	0.71	0.77	0.69	**0.85**	0.84	0.44	0.69	0.15	0.47	0.29	0.69	0.67	0.67	0.61
	0.05	0.73^{-}	0.72	0.70	0.43	0.57	0.57	**0.76**	**0.76**	0.31	0.52	0.14	0.36	0.28	0.63	0.63	0.37	0.39
	0.10	**0.64**$^{+}$	**0.64**	0.61	0.34	0.44	0.55	0.63	0.62	0.27	0.38	0.12	0.29	0.27	0.57	0.58	0.29	0.30
equal k, unequal s, k=8	0.00	**0.60**$^{+}$	0.59	0.49	0.53	0.59	0.49	0.59	0.58	0.41	0.43	0.14	0.32	0.26	0.53	0.52	0.52	0.53
	0.05	**0.55**$^{+}$	0.55	0.48	0.31	0.53	0.47	0.38	0.38	0.29	0.30	0.13	0.24	0.25	0.42	0.39	0.28	0.29
	0.10	**0.49**$^{+}$	0.48	0.44	0.25	0.46	0.42	0.27	0.27	0.24	0.24	0.12	0.18	0.24	0.37	0.37	0.22	0.23
equal k, unequal s, k=16	0.00	0.81^{-}	0.81	0.78	0.74	0.79	0.74	**0.84**	**0.84**	0.66	0.78	0.20	0.46	0.44	0.73	0.71	0.70	0.64
	0.05	**0.75**$^{+}$	**0.75**	0.73	0.48	0.62	0.63	0.71	0.71	0.58	0.67	0.18	0.40	0.43	0.71	0.68	0.45	0.43
	0.10	**0.69**$^{=}$	**0.69**	0.65	0.38	0.52	0.60	0.58	0.56	0.52	0.57	0.15	0.34	0.40	0.68	0.64	0.35	0.36
equal s, unequal k	0.00	0.77^{-}	0.77	0.73	0.71	0.71	0.56	**0.78**	0.76	0.50	0.65	0.17	0.52	0.30	0.64	0.61	0.67	0.62
	0.05	**0.66**$^{=}$	**0.66**	0.65	0.45	0.56	0.55	**0.66**	**0.66**	0.40	0.56	0.15	0.43	0.28	0.57	0.55	0.40	0.41
	0.10	0.57^{-}	0.57	0.55	0.38	0.45	0.51	0.57	**0.59**	0.35	0.51	0.14	0.36	0.26	0.53	0.48	0.33	0.35
two worlds	0.00	0.67^{-}	0.66	0.59	0.58	0.51	0.36	0.53	**0.70**	0.39	0.42	0.14	0.21	0.26	0.52	0.47	0.58	0.54
	0.05	0.50$^{=}$	**0.51**	0.46	0.29	0.42	0.34	0.33	**0.50**	0.30	0.33	0.12	0.18	0.23	0.46	0.43	0.27	0.29
	0.10	0.33^{-}	0.33	0.32	0.24	0.32	0.28	0.24	0.35	0.25	0.27	0.10	0.16	0.21	**0.40**	0.37	0.22	0.24
three big, many small	0.00	**0.66**$^{+}$	0.65	0.58	0.58	0.44	0.33	0.47	0.56	0.35	0.31	0.13	0.18	0.24	0.48	0.45	0.57	0.52
	0.05	0.40$^{=}$	**0.41**	0.38	0.23	0.35	0.29	0.29	**0.41**	0.27	0.25	0.12	0.15	0.21	0.34	0.37	0.21	0.23
	0.10	0.25^{-}	0.25	0.24	0.17	0.26	0.20	0.20	**0.30**	0.23	0.21	0.11	0.13	0.19	0.29	0.28	0.16	0.20
gradient	0.00	0.40^{-}	0.37	0.37	0.38	**0.55**	0.47	0.41	0.38	0.25	0.26	0.13	0.26	0.18	0.47	0.40	0.43	0.38
	0.05	0.28^{-}	0.28	0.24	0.23	**0.40**	0.29	0.25	0.28	0.19	0.21	0.12	0.19	0.17	0.28	0.35	0.21	0.21
	0.10	0.20^{-}	0.20	0.19	0.17	**0.31**	0.26	0.17	0.21	0.16	0.18	0.12	0.16	0.16	0.24	0.29	0.17	0.17
uniform k=8, α=1	0.05	**0.57**$^{+}$	0.56	0.50	0.35	0.55	0.49	0.43	0.43	0.34	0.35	0.14	0.25	0.30	0.47	0.40	0.33	0.35
uniform k=8, α=2	0.05	**0.58**$^{+}$	0.57	0.50	0.37	0.56	0.49	0.43	0.44	0.33	0.34	0.14	0.25	0.30	0.47	0.42	0.35	0.36
two worlds, α=1	0.05	**0.57**$^{+}$	**0.57**	0.49	0.39	0.49	0.32	0.37	0.51	0.32	0.30	0.12	0.20	0.24	0.46	0.40	0.35	0.37
two worlds, α=2	0.05	**0.65**$^{+}$	**0.65**	0.51	0.42	0.54	0.31	0.41	0.51	0.35	0.27	0.13	0.21	0.24	0.43	0.39	0.43	0.45
LFR, o_n=0		0.88^{-}	0.90	0.89	0.68	0.90	0.79	0.95	0.95	**1.00**	0.99	0.66	0.99	0.93	0.76	0.82	0.60	0.56
LFR, o_n=0.1		0.76^{-}	0.79	0.79	0.39	0.62	0.75	0.67	0.68	**0.91**	0.85	0.53	0.90	0.84	0.74	0.79	0.44	0.39
LFR, o_n=0.3		0.58^{-}	0.62	0.59	0.24	0.48	0.65	0.43	0.43	**0.75**	0.67	0.34	0.52	0.67	0.72	0.71	0.29	0.24
LFR, $\langle k \rangle$=10		0.55^{-}	0.57	0.52	0.22	0.52	0.52	0.60	0.63	**0.77**	0.66	0.41	0.66	0.57	0.50	0.39	0.25	0.25
LFR, o_m=4		0.69^{-}	0.71	0.70	0.50	0.58	0.67	0.53	0.53	**0.77**	0.67	0.43	0.70	0.72	0.72	0.73	0.49	0.45
LFR, sizes 50–250		0.54^{-}	0.53	0.55	0.25	0.51	0.34	**0.91**	0.90	0.81	0.81	0.43	0.84	0.67	0.50	0.64	0.28	0.21
LFR, C_{LFR}=0.3		0.40^{-}	0.41	0.40	0.29	0.34	0.49	0.42	0.42	0.58	0.59	0.51	0.36	**0.62**	0.49	0.43	0.31	0.33
LFR, C_{LFR}=0.5		0.32^{-}	0.32	0.31	0.28	0.33	0.33	0.27	0.28	0.33	0.36	0.43	0.24	**0.44**	0.41	0.33	0.28	0.30

TABLE S10. Per-community recovery of the true communities no larger than the median size, oracle covers, on the unequal-ring families (equal s with unequal k has no community larger than the median). Numbers of rings no larger than the median size: 20, 20, 20, 20, 16, 5 (families in order). Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
equal k, unequal s, k=8	0.00	0.62	0.62	0.52	0.57	0.62	0.52	**0.68**	**0.67**	0.48	0.49	0.15	0.33	0.39	0.57	0.55	0.55	0.55
	0.05	**0.58**	**0.58**	0.51	0.35	0.57	0.50	0.53	0.53	0.35	0.35	0.14	0.29	0.37	0.48	0.42	0.32	0.33
	0.10	**0.52**	0.51	0.47	0.28	**0.52**	0.45	0.41	0.41	0.29	0.28	0.13	0.23	0.35	0.44	0.41	0.26	0.26
equal k, unequal s, k=16	0.00	0.82	0.82	0.80	0.76	0.82	0.75	**0.91**	0.90	0.78	0.83	0.23	0.52	0.59	0.77	0.74	0.71	0.65
	0.05	0.78	0.78	0.76	0.54	0.74	0.66	**0.83**	0.81	0.74	0.73	0.19	0.48	0.57	0.72	0.71	0.48	0.48
	0.10	0.72	0.72	0.69	0.44	0.65	0.63	**0.73**	0.72	0.66	0.66	0.17	0.42	0.54	0.67	0.67	0.38	0.39
equal s, unequal k	0.00	0.77	0.77	0.73	0.72	0.75	0.57	**0.80**	0.78	0.50	0.65	0.17	0.53	0.33	0.60	0.61	0.68	0.62
	0.05	0.68	0.69	0.66	0.47	0.61	0.55	**0.70**	0.68	0.40	0.56	0.15	0.45	0.30	0.48	0.55	0.41	0.42
	0.10	0.58	0.58	0.56	0.40	0.52	0.51	**0.61**	**0.61**	0.35	0.51	0.14	0.37	0.28	0.42	0.47	0.33	0.35
two worlds	0.00	0.61	0.60	0.52	0.56	0.56	0.27	0.47	**0.68**	0.39	0.38	0.13	0.19	0.30	0.45	0.41	0.54	0.50
	0.05	0.38	0.39	0.35	0.20	0.30	0.20	0.26	**0.54**	0.30	0.28	0.11	0.15	0.26	0.31	0.33	0.20	0.22
	0.10	0.21	0.21	0.19	0.16	0.21	0.14	0.15	**0.38**	0.25	0.23	0.09	0.13	0.23	0.27	0.24	0.15	0.18
three big, many small	0.00	0.60	0.59	0.53	0.57	0.56	0.26	0.22	**0.69**	0.34	0.32	0.12	0.10	0.30	0.42	0.38	0.53	0.50
	0.05	0.33	0.34	0.31	0.19	0.27	0.20	0.11	**0.53**	0.28	0.25	0.11	0.15	0.26	0.30	0.28	0.19	0.22
	0.10	0.17	0.17	0.16	0.16	0.19	0.14	0.14	**0.37**	0.24	0.21	0.09	0.12	0.22	0.27	0.21	0.15	0.17
gradient	0.00	**0.59**	**0.59**	0.50	0.56	0.57	0.49	0.32	0.52	0.35	0.37	0.13	0.18	0.36	0.49	0.42	0.53	0.52
	0.05	**0.48**	**0.48**	0.42	0.29	0.41	0.32	0.27	0.29	0.28	0.29	0.12	0.19	0.34	0.41	0.37	0.24	0.27
	0.10	0.35	0.35	0.31	0.22	0.32	0.30	0.22	0.20	0.25	0.24	0.11	0.17	0.30	**0.36**	0.31	0.19	0.20

TABLE S11. Per-community recovery of the true communities larger than the median size, oracle covers, on the unequal-ring families (equal s with unequal k has no community larger than the median). Numbers of rings larger than the median size: 10, 10, 5, 3, 5 (families in order). Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
equal k, unequal s, k=8	0.00	0.56	0.55	0.44	0.50	0.55	0.44	**0.57**	**0.57**	0.28	0.32	0.11	0.32	0.22	0.49	0.47	0.48	0.49
	0.05	**0.50**	0.49	0.43	0.25	0.49	0.41	0.32	0.31	0.19	0.20	0.10	0.21	0.21	0.40	0.31	0.22	0.24
	0.10	**0.43**	0.42	0.38	0.20	0.42	0.36	0.24	0.23	0.17	0.17	0.10	0.16	0.20	0.36	0.29	0.17	0.18
equal k, unequal s, k=16	0.00	0.79	0.79	0.76	0.72	0.79	0.70	**0.85**	**0.86**	0.45	0.69	0.15	0.44	0.33	0.71	0.67	0.67	0.61
	0.05	0.74	0.74	0.71	0.46	0.69	0.58	**0.76**	**0.76**	0.30	0.53	0.14	0.36	0.32	0.63	0.64	0.39	0.40
	0.10	**0.66**	**0.66**	0.64	0.35	0.58	0.56	0.62	0.62	0.27	0.40	0.13	0.30	0.31	0.56	0.59	0.30	0.31
two worlds	0.00	**0.88**	**0.88**	0.87	0.81	0.87	0.77	0.73	0.87	0.37	0.62	0.17	0.50	0.27	0.65	0.71	0.78	0.70
	0.05	**0.85**	**0.85**	0.84	0.67	0.81	0.76	0.68	0.83	0.26	0.51	0.16	0.41	0.25	0.55	0.64	0.58	0.56
	0.10	**0.81**	**0.81**	0.80	0.58	0.72	0.74	0.60	0.79	0.24	0.42	0.16	0.36	0.24	0.51	0.55	0.50	0.51
three big, many small	0.00	**0.87**	**0.87**	0.86	0.80	0.86	0.78	0.58	0.82	0.23	0.28	0.18	0.45	0.17	0.63	0.70	0.78	0.71
	0.05	**0.84**	**0.84**	0.83	0.60	0.77	0.76	0.47	0.72	0.15	0.26	0.19	0.29	0.17	0.55	0.58	0.52	0.54
	0.10	0.77	**0.77**	0.74	0.53	0.68	0.72	0.31	0.47	0.13	0.23	0.20	0.25	0.17	0.47	0.51	0.44	0.48
gradient	0.00	**0.60**	0.59	0.52	0.54	0.59	0.50	0.40	0.44	0.17	0.15	0.12	0.28	0.11	0.50	0.44	0.51	0.49
	0.05	**0.49**	**0.49**	0.43	0.25	0.42	0.32	0.20	0.26	0.11	0.13	0.13	0.19	0.11	0.37	0.38	0.20	0.21
	0.10	**0.37**	**0.36**	0.33	0.22	0.33	0.31	0.15	0.22	0.09	0.12	0.14	0.17	0.10	0.31	0.31	0.18	0.17

TABLE S12. Coverage (fraction of nodes in at least one community) of the oracle covers.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.94	0.95	0.79	0.79	0.95	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.71	0.71	0.78	0.74
	0.05	0.95	0.95	0.80	0.74	0.95	0.80	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.72	0.71	0.71	0.66
	0.10	0.94	0.95	0.81	0.68	0.96	0.81	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.73	0.72	0.66	0.60
uniform, k=16	0.00	0.95	0.95	0.96	0.84	0.99	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.91	0.82	0.76
	0.05	0.93	0.94	0.96	0.76	0.99	0.93	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.92	0.71	0.69
	0.10	0.91	0.91	0.96	0.70	0.99	0.91	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.92	0.66	0.65
equal k, unequal s, k=8	0.00	0.95	0.96	0.79	0.80	0.96	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.71	0.71	0.79	0.74
	0.05	0.95	0.95	0.80	0.74	0.96	0.80	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.72	0.71	0.71	0.66
	0.10	0.95	0.95	0.81	0.69	0.96	0.81	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.73	0.72	0.66	0.60
equal k, unequal s, k=16	0.00	0.95	0.95	0.96	0.85	0.99	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.91	0.82	0.74
	0.05	0.93	0.93	0.96	0.76	0.99	0.94	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.92	0.72	0.68
	0.10	0.91	0.92	0.96	0.70	0.99	0.90	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.94	0.92	0.65	0.65
equal s, unequal k	0.00	0.92	0.93	0.94	0.85	0.97	0.83	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.93	0.91	0.81	0.79
	0.05	0.91	0.91	0.93	0.72	0.96	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.92	0.67	0.70
	0.10	0.87	0.87	0.89	0.67	0.96	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.95	0.92	0.63	0.66
two worlds	0.00	0.92	0.93	0.91	0.83	0.96	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.96	0.90	0.87	0.80	0.78
	0.05	0.87	0.89	0.87	0.66	0.93	0.75	1.00	1.00	1.00	1.00	1.00	1.00	0.96	0.91	0.88	0.64	0.68
	0.10	0.83	0.84	0.82	0.61	0.92	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.93	0.88	0.60	0.67
three big, many small	0.00	0.94	0.94	0.93	0.84	0.97	0.83	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.92	0.90	0.81	0.81
	0.05	0.88	0.89	0.89	0.71	0.95	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.94	0.91	0.68	0.73
	0.10	0.83	0.84	0.84	0.72	0.94	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.95	0.92	0.66	0.72
gradient	0.00	0.97	0.97	0.89	0.81	0.98	0.89	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.83	0.82	0.73	0.68
	0.05	0.96	0.96	0.90	0.79	0.98	0.89	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.84	0.82	0.70	0.63
	0.10	0.95	0.96	0.90	0.77	0.98	0.89	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.83	0.75	0.61
uniform k=8, α=1	0.05	0.94	0.96	0.81	0.75	0.96	0.81	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.72	0.71	0.72	0.67
uniform k=8, α=2	0.05	0.94	0.95	0.81	0.75	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.73	0.72	0.74	0.70
two worlds, α=1	0.05	0.93	0.94	0.89	0.75	0.96	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.89	0.86	0.70	0.69
two worlds, α=2	0.05	0.93	0.93	0.88	0.79	0.98	0.70	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.89	0.86	0.76	0.77
LFR, o_n=0		0.97	0.97	0.96	0.80	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.98	0.97	0.72	0.55
LFR, o_n=0.1		0.94	0.95	0.93	0.65	0.99	0.77	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.96	0.61	0.48
LFR, o_n=0.3		0.87	0.90	0.88	0.48	0.97	0.73	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.97	0.95	0.49	0.34
LFR, $\langle k \rangle$=10		0.89	0.90	0.70	0.47	0.91	0.70	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.59	0.57	0.47	0.37
LFR, o_m=4		0.94	0.95	0.93	0.77	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.98	0.96	0.73	0.56
LFR, sizes 50–250		0.80	0.83	0.84	0.45	0.96	0.42	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.91	0.86	0.39	0.33
LFR, C_{LFR}=0.3		0.82	0.82	0.82	0.63	0.96	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.58	0.59
LFR, C_{LFR}=0.5		0.88	0.88	0.88	0.82	0.99	0.86	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.77	0.78

TABLE S13. Number of communities of the oracle covers (planted: Table VIII).

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	190	203	172	92	241	172	72	72	106	88	111	40	262	82	71	95	91
	0.05	196	212	176	111	281	175	71	71	114	89	86	41	258	113	108	117	107
	0.10	204	223	180	115	338	176	72	71	115	93	74	46	253	135	125	114	109
uniform, k=16	0.00	101	105	163	42	431	178	27	27	132	33	50	18	140	82	88	49	45
	0.05	95	99	169	59	529	215	28	29	144	38	34	23	135	146	110	68	57
	0.10	90	94	177	68	662	221	30	31	169	43	26	24	125	186	123	70	63
equal k, unequal s, k=8	0.00	187	199	170	86	231	170	60	61	102	90	114	26	265	75	68	92	87
	0.05	194	209	172	106	273	171	62	65	110	93	91	33	258	103	103	114	106
	0.10	199	216	177	111	322	176	65	67	110	96	72	37	254	124	120	112	106
equal k, unequal s, k=16	0.00	107	110	169	51	439	184	37	36	131	39	49	31	135	93	95	54	52
	0.05	101	105	176	63	524	220	37	37	139	40	35	30	130	151	115	66	61
	0.10	94	98	184	71	650	225	37	38	163	41	27	31	121	205	131	74	66
equal s, unequal k	0.00	80	88	146	48	697	186	24	34	155	40	53	16	102	233	128	48	49
	0.05	89	93	145	70	772	168	24	39	189	41	38	18	90	455	152	65	62
	0.10	100	105	148	70	892	168	24	41	199	41	30	19	79	542	243	64	61
two worlds	0.00	105	114	145	57	774	188	19	36	183	59	60	17	108	250	122	56	55
	0.05	124	133	156	81	855	176	14	39	235	57	44	15	94	465	219	71	68
	0.10	136	143	162	73	957	159	12	40	229	55	31	16	81	481	393	62	63
three big, many small	0.00	92	97	147	45	884	193	9	24	195	50	51	8	96	261	95	45	46
	0.05	106	110	158	62	931	185	8	25	264	46	37	14	85	393	190	57	54
	0.10	108	115	157	56	1032	173	13	27	255	41	29	15	71	476	271	48	49
gradient	0.00	140	156	155	53	344	159	22	14	118	76	79	12	209	56	53	51	48
	0.05	147	162	165	66	450	184	28	15	127	73	60	18	201	81	57	67	61
	0.10	156	170	183	63	539	193	30	16	151	71	49	22	190	95	72	55	57
uniform k=8, α=1	0.05	206	226	180	106	319	179	69	73	116	104	88	44	257	112	114	117	106
uniform k=8, α=2	0.05	209	220	189	105	364	188	72	74	117	114	88	43	257	114	117	117	110
two worlds, α=1	0.05	109	114	137	70	972	140	16	56	204	198	45	16	105	651	413	75	68
two worlds, α=2	0.05	82	85	116	49	1085	106	17	72	61	316	48	16	105	986	476	63	60
LFR, o_n=0		83	89	99	65	751	107	40	40	40	40	27	40	41	602	75	48	46
LFR, o_n=0.1		91	97	106	65	868	113	44	44	44	44	26	51	47	746	90	49	44
LFR, o_n=0.3		107	118	128	64	1156	124	52	52	94	48	22	65	61	946	138	49	42
LFR, $\langle k \rangle$=10		255	284	241	105	481	240	46	48	62	62	24	33	204	99	101	90	89
LFR, o_m=4		89	98	101	67	1060	103	52	52	53	53	29	68	73	378	103	57	56
LFR, sizes 50–250		165	183	244	71	1236	151	18	20	110	20	8	21	18	358	161	47	41
LFR, C_{LFR}=0.3		119	123	123	90	1742	132	43	50	58	42	31	66	88	1950	1401	64	65
LFR, C_{LFR}=0.5		120	121	120	114	1496	123	57	69	78	47	72	57	121	1143	430	101	106

TABLE S14. Coverage (fraction of nodes in at least one community) of the label-free covers.

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.96	0.97	0.79	0.82	0.97	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.71	0.71	0.80	0.74
	0.05	0.96	0.97	0.80	0.77	0.97	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.72	0.71	0.73	0.68
	0.10	0.96	0.96	0.81	0.75	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.73	0.72	0.68	0.63
uniform, k=16	0.00	0.95	0.95	0.96	0.86	0.97	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.91	0.84	0.79
	0.05	0.94	0.94	0.95	0.78	0.97	0.93	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.92	0.75	0.71
	0.10	0.92	0.92	0.94	0.75	0.97	0.92	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.92	0.69	0.67
equal k, unequal s, k=8	0.00	0.96	0.97	0.79	0.82	0.97	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.71	0.71	0.80	0.74
	0.05	0.96	0.96	0.80	0.76	0.97	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.72	0.71	0.73	0.68
	0.10	0.96	0.96	0.81	0.74	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.73	0.72	0.68	0.63
equal k, unequal s, k=16	0.00	0.95	0.95	0.96	0.86	0.97	0.97	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.91	0.84	0.80
	0.05	0.94	0.94	0.95	0.80	0.97	0.95	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.92	0.76	0.72
	0.10	0.92	0.92	0.95	0.75	0.97	0.91	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.92	0.70	0.67
equal s, unequal k	0.00	0.93	0.93	0.94	0.86	0.96	0.85	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.91	0.83	0.83
	0.05	0.92	0.92	0.93	0.77	0.95	0.84	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.92	0.71	0.72
	0.10	0.88	0.90	0.90	0.71	0.94	0.82	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.95	0.93	0.64	0.67
two worlds	0.00	0.93	0.93	0.90	0.85	0.92	0.75	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.90	0.87	0.80	0.79
	0.05	0.89	0.89	0.87	0.67	0.92	0.74	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.91	0.88	0.63	0.66
	0.10	0.81	0.82	0.81	0.62	0.91	0.70	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.93	0.89	0.57	0.63
three big, many small	0.00	0.92	0.92	0.90	0.85	0.94	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.92	0.90	0.81	0.78
	0.05	0.84	0.86	0.85	0.64	0.93	0.75	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.94	0.91	0.60	0.63
	0.10	0.77	0.78	0.77	0.60	0.91	0.68	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.95	0.91	0.55	0.64
gradient	0.00	0.93	0.93	0.87	0.82	0.95	0.88	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.83	0.82	0.80	0.74
	0.05	0.91	0.92	0.86	0.76	0.95	0.86	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.84	0.82	0.72	0.70
	0.10	0.90	0.91	0.86	0.72	0.95	0.84	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.85	0.83	0.71	0.66
uniform k=8, α=1	0.05	0.97	0.97	0.80	0.78	0.98	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.72	0.71	0.75	0.68
uniform k=8, α=2	0.05	0.97	0.97	0.81	0.80	0.98	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.73	0.72	0.78	0.71
two worlds, α=1	0.05	0.91	0.92	0.86	0.82	0.94	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.89	0.87	0.75	0.74
two worlds, α=2	0.05	0.93	0.93	0.86	0.88	0.95	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.89	0.87	0.80	0.79
LFR, o_n=0		0.97	0.97	0.97	0.87	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.97	0.86	0.65
LFR, o_n=0.1		0.96	0.96	0.95	0.75	0.99	0.77	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.96	0.70	0.61
LFR, o_n=0.3		0.91	0.91	0.91	0.58	0.98	0.73	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.95	0.55	0.52
LFR, $\langle k \rangle$=10		0.90	0.91	0.69	0.58	0.95	0.70	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.59	0.57	0.53	0.39
LFR, o_m=4		0.95	0.95	0.94	0.82	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.96	0.81	0.68
LFR, sizes 50–250		0.86	0.87	0.86	0.54	0.94	0.60	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.91	0.86	0.47	0.45
LFR, C_{LFR}=0.3		0.88	0.88	0.88	0.71	0.99	0.84	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.67	0.65
LFR, C_{LFR}=0.5		0.93	0.93	0.93	0.87	1.00	0.92	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.84	0.84

TABLE S15. Number of communities of the label-free covers (planted: Table VIII).

		HC and ablations				relatives		partitions						overlapping methods				
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	190	201	172	80	220	172	30	31	98	88	111	23	237	83	67	89	90
	0.05	192	206	174	90	238	174	25	25	95	89	86	17	229	120	79	100	99
	0.10	197	214	176	90	265	175	21	23	96	93	74	18	222	142	107	109	98
uniform, k=16	0.00	101	105	151	40	372	176	19	19	131	33	50	17	116	94	77	45	44
	0.05	94	97	153	52	422	215	19	19	140	38	34	18	111	141	97	57	57
	0.10	88	92	155	57	499	223	19	19	164	43	26	19	105	175	121	63	61
equal k, unequal s, k=8	0.00	186	197	170	79	214	170	26	27	100	90	114	22	236	80	67	86	86
	0.05	193	206	171	92	239	171	24	25	99	93	91	19	228	120	88	105	97
	0.10	196	213	175	92	266	174	21	23	96	96	72	18	221	140	104	106	97
equal k, unequal s, k=16	0.00	107	110	155	47	377	182	27	27	130	39	49	22	115	97	82	51	50
	0.05	100	103	160	51	429	218	25	25	138	40	35	22	112	138	105	56	58
	0.10	91	95	162	59	491	226	22	22	162	41	27	20	104	166	124	65	63
equal s, unequal k	0.00	83	89	138	45	631	196	18	27	153	40	53	15	89	129	128	45	46
	0.05	90	95	138	64	688	188	17	31	187	41	38	14	80	165	142	62	59
	0.10	100	107	141	67	764	176	17	34	198	41	30	14	70	193	152	62	61
two worlds	0.00	102	107	131	49	666	157	21	32	178	59	60	13	102	152	138	56	55
	0.05	117	121	140	81	744	156	18	43	225	57	44	14	89	181	150	72	69
	0.10	129	134	147	73	828	143	16	52	226	55	31	13	76	211	184	65	65
three big, many small	0.00	79	84	110	42	776	160	21	36	187	50	51	15	89	180	136	44	47
	0.05	96	102	119	65	851	145	18	40	239	46	37	16	78	289	225	57	55
	0.10	97	104	116	57	932	114	17	43	248	41	29	15	67	303	322	50	50
gradient	0.00	167	185	171	68	321	164	24	30	122	76	79	19	172	79	82	61	59
	0.05	181	194	190	84	413	199	22	29	125	73	60	19	165	131	100	82	76
	0.10	188	199	200	89	506	212	21	27	140	71	49	19	158	162	134	82	77
uniform k=8, α=1	0.05	207	221	178	87	270	179	25	26	98	104	88	20	229	120	79	102	100
uniform k=8, α=2	0.05	203	214	184	82	307	187	25	29	104	114	88	17	228	125	85	98	99
two worlds, α=1	0.05	106	110	121	59	849	134	19	66	246	198	45	14	94	181	103	71	65
two worlds, α=2	0.05	81	84	114	33	911	110	21	82	192	316	48	15	90	126	110	57	55
LFR, o_n=0		82	88	97	54	749	107	38	39	40	40	27	40	41	141	72	39	43
LFR, o_n=0.1		86	93	105	56	787	113	32	32	44	44	26	51	43	169	84	46	41
LFR, o_n=0.3		100	113	123	59	1005	124	26	26	93	48	22	37	49	214	114	47	41
LFR, $\langle k \rangle$=10		244	275	233	92	405	240	30	33	62	62	24	33	198	82	47	86	89
LFR, o_m=4		89	97	100	56	935	103	33	34	52	53	29	68	54	173	94	48	53
LFR, sizes 50–250		184	197	235	70	1046	272	18	19	110	20	8	21	15	213	161	48	45
LFR, C_{LFR}=0.3		105	108	107	78	1258	141	23	24	53	42	31	29	82	261	464	68	63
LFR, C_{LFR}=0.5		90	92	90	82	985	124	22	26	36	47	72	23	109	187	205	79	85

TABLE S16. The paired intervals behind Table I: Ω of HyperCommon minus that of the best (first line of each cell) and second-best other method, with 95% bootstrap intervals over the ten test seeds, ablations excluded.

Family	o_n	oracle	label-free	frozen	native
uniform, k=8	0.00	Leiden −0.054 [−0.057, −0.051]	Leiden −0.001 [−0.009, +0.008]	Leiden −0.080 [−0.084, −0.078]	Leiden −0.003 [−0.010, +0.003]
		L-CPM −0.050 [−0.053, −0.048]	L-CPM +0.002 [−0.004, +0.007]	L-CPM −0.080 [−0.082, −0.078]	L-CPM +0.004 [−0.000, +0.009]
	0.05	Link +0.005 [+0.002, +0.008]	Link +0.024 [+0.018, +0.030]	Leiden −0.026 [−0.034, −0.019]	CPM +0.063 [+0.058, +0.068]
		Leiden +0.020 [+0.016, +0.025]	CPM +0.063 [+0.058, +0.068]	L-CPM −0.020 [−0.026, −0.014]	Leiden +0.103 [+0.097, +0.108]
	0.10	Link +0.009 [+0.004, +0.013]	Link +0.049 [+0.040, +0.057]	CPM +0.005 [−0.001, +0.010]	CPM +0.063 [+0.054, +0.072]
		CPM +0.079 [+0.071, +0.086]	CPM +0.063 [+0.054, +0.072]	Link +0.008 [+0.000, +0.014]	ANGEL +0.095 [+0.087, +0.102]
uniform, k=16	0.00	Leiden −0.078 [−0.082, −0.074]	Leiden −0.065 [−0.070, −0.060]	Leiden −0.101 [−0.106, −0.096]	Leiden −0.065 [−0.068, −0.061]
		L-CPM −0.073 [−0.076, −0.068]	L-CPM −0.058 [−0.062, −0.054]	L-CPM −0.097 [−0.103, −0.092]	L-CPM −0.059 [−0.063, −0.055]
	0.05	Leiden −0.044 [−0.049, −0.039]	Leiden −0.035 [−0.040, −0.029]	Leiden −0.193 [−0.208, −0.177]	Leiden −0.037 [−0.043, −0.030]
		L-CPM −0.040 [−0.045, −0.035]	L-CPM −0.029 [−0.037, −0.022]	L-CPM −0.186 [−0.200, −0.169]	L-CPM −0.029 [−0.036, −0.021]
	0.10	Leiden −0.008 [−0.017, +0.000]	Leiden +0.002 [−0.009, +0.011]	Leiden −0.316 [−0.328, −0.305]	Leiden +0.003 [−0.007, +0.010]
		L-CPM +0.001 [−0.009, +0.008]	L-CPM +0.016 [+0.006, +0.025]	L-CPM −0.308 [−0.320, −0.296]	L-CPM +0.012 [+0.002, +0.020]
equal k, unequal s, k=8	0.00	Leiden −0.028 [−0.031, −0.024]	Leiden +0.012 [+0.008, +0.016]	Leiden −0.044 [−0.057, −0.032]	Leiden +0.010 [+0.008, +0.013]
		L-CPM −0.025 [−0.028, −0.022]	Link +0.013 [+0.008, +0.018]	L-CPM −0.041 [−0.052, −0.031]	L-CPM +0.013 [+0.010, +0.017]
	0.05	Link +0.008 [+0.003, +0.012]	Link +0.029 [+0.023, +0.035]	Link +0.000 [−0.004, +0.006]	CPM +0.093 [+0.086, +0.100]
		CPM +0.095 [+0.088, +0.101]	CPM +0.093 [+0.085, +0.100]	CPM +0.033 [+0.028, +0.038]	DEMON +0.144 [+0.138, +0.149]
	0.10	Link +0.002 [−0.003, +0.008]	Link +0.038 [+0.029, +0.048]	CPM +0.000 [−0.010, +0.013]	CPM +0.084 [+0.073, +0.095]
		CPM +0.084 [+0.073, +0.097]	CPM +0.084 [+0.073, +0.096]	Link +0.003 [−0.004, +0.012]	DEMON +0.118 [+0.111, +0.126]
equal k, unequal s, k=16	0.00	Leiden −0.074 [−0.078, −0.071]	Leiden −0.045 [−0.048, −0.041]	Leiden −0.101 [−0.104, −0.096]	Leiden −0.044 [−0.048, −0.040]
		L-CPM −0.072 [−0.074, −0.068]	L-CPM −0.040 [−0.045, −0.035]	L-CPM −0.093 [−0.097, −0.091]	L-CPM −0.040 [−0.044, −0.036]
	0.05	Leiden −0.039 [−0.042, −0.036]	Leiden +0.004 [−0.004, +0.010]	Leiden −0.173 [−0.185, −0.161]	Leiden +0.009 [+0.002, +0.017]
		L-CPM −0.030 [−0.034, −0.026]	L-CPM +0.007 [−0.001, +0.014]	L-CPM −0.162 [−0.175, −0.148]	L-CPM +0.015 [+0.007, +0.022]
	0.10	Leiden +0.010 [+0.001, +0.018]	ANGEL +0.039 [+0.028, +0.049]	Leiden −0.306 [−0.320, −0.291]	ANGEL +0.039 [+0.027, +0.050]
		L-CPM +0.016 [+0.008, +0.023]	Leiden +0.055 [+0.043, +0.067]	L-CPM −0.303 [−0.316, −0.289]	Leiden +0.059 [+0.049, +0.069]
equal s, unequal k	0.00	Leiden −0.028 [−0.031, −0.025]	L-CPM −0.005 [−0.012, +0.002]	Leiden −0.070 [−0.073, −0.067]	Leiden +0.000 [−0.006, +0.006]
		L-CPM −0.027 [−0.030, −0.024]	Leiden −0.002 [−0.008, +0.004]	L-CPM −0.036 [−0.039, −0.033]	L-CPM +0.033 [+0.028, +0.037]
	0.05	L-CPM −0.011 [−0.016, −0.005]	L-CPM −0.006 [−0.012, +0.001]	Leiden −0.183 [−0.202, −0.165]	L-CPM +0.025 [+0.017, +0.033]
		Leiden −0.006 [−0.014, +0.001]	Leiden +0.021 [+0.013, +0.030]	L-CPM −0.179 [−0.195, −0.165]	Leiden +0.027 [+0.019, +0.034]
	0.10	L-CPM −0.029 [−0.034, −0.023]	L-CPM −0.017 [−0.024, −0.010]	L-CPM −0.328 [−0.337, −0.319]	L-CPM +0.028 [+0.020, +0.037]
		Leiden −0.006 [−0.013, +0.001]	Leiden +0.031 [+0.024, +0.039]	Leiden −0.316 [−0.325, −0.306]	Leiden +0.031 [+0.025, +0.039]
two worlds	0.00	L-CPM −0.004 [−0.007, −0.000]	L-CPM +0.035 [+0.027, +0.042]	L-CPM −0.051 [−0.056, −0.046]	L-CPM +0.007 [+0.002, +0.011]
		Link +0.019 [+0.013, +0.027]	Link +0.042 [+0.034, +0.049]	Link −0.001 [−0.004, +0.003]	SCAN +0.104 [+0.090, +0.119]
	0.05	L-CPM −0.007 [−0.017, −0.000]	CPM +0.100 [+0.079, +0.117]	L-CPM −0.249 [−0.269, −0.230]	L-CPM +0.001 [−0.012, +0.014]
		Link +0.065 [+0.053, +0.077]	Link +0.103 [+0.086, +0.120]	CPM −0.121 [−0.139, −0.102]	CPM +0.100 [+0.079, +0.117]
	0.10	L-CPM −0.004 [−0.012, +0.004]	CPM +0.063 [+0.052, +0.074]	L-CPM −0.383 [−0.398, −0.369]	L-CPM +0.009 [−0.001, +0.019]
		CPM +0.061 [+0.050, +0.071]	Link +0.184 [+0.165, +0.203]	CPM −0.189 [−0.206, −0.172]	CPM +0.063 [+0.052, +0.074]
three big, many small	0.00	Link +0.009 [+0.005, +0.017]	Link +0.025 [+0.019, +0.035]	Link +0.026 [+0.014, +0.036]	SCAN +0.092 [+0.081, +0.103]
		L-CPM +0.032 [+0.025, +0.039]	SCAN +0.092 [+0.081, +0.102]	CPM +0.100 [+0.088, +0.111]	CPM +0.112 [+0.103, +0.120]
	0.05	Link +0.075 [+0.059, +0.090]	CPM +0.102 [+0.084, +0.122]	CPM −0.159 [−0.177, −0.141]	CPM +0.102 [+0.083, +0.123]
		CPM +0.084 [+0.073, +0.097]	Link +0.103 [+0.077, +0.125]	L-CPM −0.108 [−0.131, −0.081]	ANGEL +0.264 [+0.249, +0.278]
	0.10	CPM +0.043 [+0.026, +0.059]	CPM +0.059 [+0.033, +0.086]	CPM −0.230 [−0.244, −0.217]	CPM +0.059 [+0.034, +0.086]
		Link +0.098 [+0.078, +0.118]	Link +0.122 [+0.085, +0.152]	ANGEL −0.117 [−0.129, −0.105]	ANGEL +0.239 [+0.222, +0.256]
gradient	0.00	Link −0.012 [−0.021, −0.003]	Link −0.186 [−0.210, −0.165]	Link −0.002 [−0.018, +0.013]	CPM −0.139 [−0.164, −0.115]
		CPM +0.078 [+0.064, +0.095]	CPM −0.139 [−0.165, −0.114]	CPM +0.024 [+0.009, +0.040]	DEMON −0.080 [−0.104, −0.058]
	0.05	Link +0.033 [+0.016, +0.050]	Link −0.162 [−0.177, −0.147]	ANGEL −0.022 [−0.046, +0.000]	ANGEL −0.107 [−0.125, −0.091]
		ANGEL +0.103 [+0.085, +0.119]	ANGEL −0.107 [−0.124, −0.091]	DEMON −0.003 [−0.018, +0.013]	CPM −0.087 [−0.098, −0.074]
	0.10	Link +0.028 [+0.016, +0.039]	Link −0.152 [−0.177, −0.128]	DEMON −0.136 [−0.152, −0.120]	CPM −0.116 [−0.139, −0.090]
		CPM +0.072 [+0.060, +0.084]	CPM −0.116 [−0.138, −0.086]	ANGEL −0.119 [−0.136, −0.105]	ANGEL −0.095 [−0.119, −0.069]
uniform k=8, α=1	0.05	Link +0.008 [+0.005, +0.011]	Link +0.035 [+0.027, +0.043]	Leiden −0.027 [−0.033, −0.022]	CPM +0.079 [+0.074, +0.084]
		Leiden +0.019 [+0.015, +0.023]	CPM +0.079 [+0.074, +0.084]	L-CPM −0.021 [−0.027, −0.016]	Leiden +0.116 [+0.110, +0.123]
uniform k=8, α=2	0.05	Link +0.010 [+0.007, +0.013]	Link +0.024 [+0.018, +0.031]	Leiden −0.021 [−0.027, −0.016]	CPM +0.081 [+0.076, +0.087]
		Leiden +0.020 [+0.017, +0.024]	CPM +0.081 [+0.076, +0.087]	L-CPM −0.013 [−0.020, −0.006]	Leiden +0.115 [+0.108, +0.120]
two worlds, α=1	0.05	L-CPM −0.046 [−0.051, −0.040]	L-CPM +0.115 [+0.099, +0.129]	L-CPM −0.285 [−0.304, −0.266]	CPM +0.123 [+0.111, +0.134]
		Link +0.072 [+0.065, +0.077]	Link +0.117 [+0.104, +0.129]	Walk. −0.108 [−0.125, −0.090]	Walk. +0.155 [+0.140, +0.171]
two worlds, α=2	0.05	L-CPM +0.001 [−0.007, +0.009]	L-CPM +0.131 [+0.116, +0.144]	L-CPM −0.200 [−0.218, −0.183]	CPM +0.152 [+0.137, +0.167]
		Link +0.071 [+0.064, +0.078]	Link +0.131 [+0.118, +0.145]	Leiden −0.091 [−0.110, −0.073]	Walk. +0.171 [+0.154, +0.188]
LFR, o_n=0		Leiden −0.143 [−0.156, −0.130]	Info. −0.159 [−0.172, −0.146]	Leiden −0.183 [−0.199, −0.166]	Info. −0.159 [−0.172, −0.146]
		L-CPM −0.143 [−0.156, −0.130]	SBM −0.149 [−0.161, −0.137]	Info. −0.183 [−0.200, −0.166]	SBM −0.149 [−0.161, −0.137]
LFR, o_n=0.1		L-CPM −0.148 [−0.159, −0.135]	Info. −0.185 [−0.216, −0.157]	Info. −0.243 [−0.261, −0.221]	Info. −0.190 [−0.219, −0.162]
		Leiden −0.147 [−0.159, −0.134]	SBM −0.173 [−0.201, −0.144]	Leiden −0.241 [−0.260, −0.218]	SBM −0.173 [−0.202, −0.143]
LFR, o_n=0.3		SLPA −0.209 [−0.225, −0.192]	Info. −0.190 [−0.214, −0.163]	Leiden −0.397 [−0.407, −0.386]	SLPA −0.212 [−0.242, −0.183]
		L-CPM −0.192 [−0.206, −0.179]	ANGEL −0.182 [−0.197, −0.166]	L-CPM −0.397 [−0.407, −0.386]	Info. −0.190 [−0.216, −0.166]
LFR, $\langle k \rangle$=10		Leiden −0.330 [−0.342, −0.316]	Info. −0.276 [−0.298, −0.256]	Leiden −0.365 [−0.386, −0.343]	Info. −0.281 [−0.303, −0.261]
		L-CPM −0.329 [−0.341, −0.316]	L-CPM −0.264 [−0.294, −0.233]	L-CPM −0.361 [−0.382, −0.341]	Leiden −0.230 [−0.251, −0.209]
LFR, o_m=4		SLPA −0.152 [−0.166, −0.138]	Info. −0.095 [−0.118, −0.078]	SLPA −0.184 [−0.195, −0.174]	SLPA −0.101 [−0.122, −0.084]
		L-CPM −0.094 [−0.108, −0.083]	SLPA −0.066 [−0.087, −0.049]	L-CPM −0.159 [−0.168, −0.150]	Info. −0.098 [−0.121, −0.082]
LFR, sizes 50–250		Leiden −0.448 [−0.467, −0.427]	Leiden −0.454 [−0.477, −0.431]	Leiden −0.547 [−0.572, −0.521]	Leiden −0.425 [−0.449, −0.400]
		L-CPM −0.429 [−0.447, −0.409]	L-CPM −0.434 [−0.455, −0.412]	L-CPM −0.520 [−0.542, −0.499]	L-CPM −0.383 [−0.404, −0.363]
LFR, C_{LFR}=0.3		Leiden −0.229 [−0.241, −0.215]	SLPA −0.234 [−0.255, −0.212]	Leiden −0.310 [−0.323, −0.297]	SLPA −0.254 [−0.276, −0.232]
		L-CPM −0.222 [−0.233, −0.208]	Info. −0.218 [−0.241, −0.191]	L-CPM −0.306 [−0.319, −0.294]	Info. −0.253 [−0.273, −0.231]
LFR, C_{LFR}=0.5		Leiden −0.076 [−0.082, −0.070]	LPA −0.105 [−0.114, −0.096]	SLPA −0.138 [−0.150, −0.128]	SLPA −0.118 [−0.132, −0.105]
		L-CPM −0.071 [−0.079, −0.065]	SLPA −0.104 [−0.119, −0.092]	Leiden −0.129 [−0.138, −0.120]	LPA −0.105 [−0.115, −0.096]

TABLE S17. The label-free selector against the oracle on the ring settings, for HyperCommon, link communities, clique percolation, the two Leiden objectives, Infomap and SLPA: the ratio of label-free to oracle Ω (both mean over $p \in [0, 0.8]$) and the median number of communities of the selected covers over the test graphs, label-free/oracle (number of planted rings in the first column). For a partition the selector is the Newman–Girvan modularity.

			HC		Link		CPM		Leiden		L-CPM		Infomap		SLPA	
Family	o_n	rings	ratio	comm.	ratio	comm.	ratio	comm.	ratio	comm.	ratio	comm.	ratio	comm.	ratio	comm.
uniform, k=8	0.00	40	0.98	77/80	0.97	140/144	1.00	155/155	0.91	36/41	0.91	36/40	0.99	69/132	0.76	206/250
	0.05	40	0.96	92/94	0.92	170/213	1.00	159/159	0.80	28/45	0.79	27/47	0.91	70/132	0.75	200/250
	0.10	40	0.95	108/112	0.88	196/250	0.98	172/172	0.77	23/52	0.77	24/52	0.92	126/128	0.75	198/246
uniform, k=16	0.00	20	1.00	20/20	0.97	274/292	1.00	53/64	0.98	20/20	0.98	20/20	1.00	70/71	0.78	103/128
	0.05	20	0.99	20/21	0.82	345/417	1.00	142/142	0.98	20/20	0.97	20/20	0.99	71/71	0.78	101/124
	0.10	20	0.98	21/24	0.74	490/584	1.00	140/140	0.97	20/21	0.96	20/22	0.98	72/72	0.77	98/121
equal k, unequal s, k=8	0.00	30	0.99	76/76	0.98	138/138	1.00	153/153	0.93	30/40	0.93	30/40	0.99	73/73	0.73	200/255
	0.05	30	0.99	92/93	0.95	156/160	1.00	162/162	0.88	27/46	0.88	27/50	0.97	74/131	0.72	202/254
	0.10	30	0.99	110/112	0.91	182/238	0.99	174/174	0.82	23/51	0.83	23/53	0.94	124/126	0.71	193/252
equal k, unequal s, k=16	0.00	30	1.00	30/30	0.97	292/299	1.00	66/70	0.97	30/30	0.96	30/30	1.00	55/55	0.82	104/124
	0.05	30	0.98	30/30	0.81	336/422	1.00	148/148	0.93	26/30	0.93	27/30	0.99	58/58	0.81	100/122
	0.10	30	0.97	30/32	0.71	416/594	0.99	151/151	0.90	24/31	0.89	24/31	0.99	61/61	0.79	98/116
equal s, unequal k	0.00	20	0.99	37/39	0.95	589/664	0.99	90/90	0.96	20/20	0.97	28/31	1.00	56/56	0.87	100/116
	0.05	20	0.98	58/68	0.90	674/696	1.00	94/94	0.94	19/22	0.97	31/34	0.99	57/57	0.88	94/110
	0.10	20	0.98	73/76	0.84	766/810	0.99	93/93	0.92	17/22	0.97	33/37	0.99	57/57	0.88	88/100
two worlds	0.00	25	0.99	48/48	0.96	719/828	0.99	118/118	0.87	23/18	0.94	26/26	0.99	72/74	0.87	114/128
	0.05	25	0.98	108/114	0.93	801/780	0.96	107/96	0.79	19/13	0.82	41/34	0.98	72/72	0.88	107/113
	0.10	25	0.99	114/112	0.86	830/834	0.98	95/79	0.83	17/10	0.69	53/38	0.99	69/69	0.90	94/96
three big, many small	0.00	19	0.99	40/44	0.97	746/976	0.97	94/83	0.67	22/8	0.49	34/22	0.85	61/61	0.85	101/114
	0.05	19	0.98	91/92	0.93	814/820	0.95	89/74	0.76	18/6	0.49	40/25	0.97	63/65	0.87	93/100
	0.10	19	0.94	85/86	0.89	848/880	0.91	85/68	0.89	17/12	0.62	43/28	0.99	62/62	0.89	79/80
gradient	0.00	10	0.64	108/44	0.91	216/196	0.97	134/99	0.59	26/16	0.47	30/14	0.77	97/97	0.70	147/186
	0.05	10	0.56	117/70	0.89	362/346	0.93	173/136	0.88	24/22	0.66	30/14	0.97	94/94	0.72	145/187
	0.10	10	0.49	124/87	0.86	492/414	0.86	184/138	0.89	22/24	0.64	27/14	0.96	89/90	0.74	140/182
uniform k=8, α=1	0.05	40	0.97	92/92	0.93	165/213	1.00	167/167	0.80	27/48	0.80	28/46	0.92	72/132	0.75	199/250
uniform k=8, α=2	0.05	40	0.97	91/92	0.94	197/208	1.00	214/214	0.80	28/46	0.80	30/46	0.93	132/134	0.76	199/248
two worlds, α=1	0.05	25	1.00	80/83	0.93	836/862	0.96	102/80	0.77	20/14	0.80	69/49	0.98	75/74	0.88	113/128
two worlds, α=2	0.05	25	0.99	49/52	0.91	898/1072	0.98	89/89	0.75	21/15	0.83	72/51	0.96	76/70	0.87	112/132

TABLE S18. Grid-edge audit: share of the test graphs (all ring and LFR families) whose selected parameter lies at an end of the method's grid (Table IX); for SCAN the end points of ε; in brackets the shares at the low and the high end. HyperCommon's lowest threshold, 0.02, is an end of its grid, and selecting it means the covers coalesce. The low end of clique percolation, clique size 3, is the smallest possible.

Method	oracle	label-free	frozen
HC	2% (2/0)	2% (2/0)	0% (0/0)
e-c.	2% (2/0)	2% (2/0)	0% (0/0)
tri.	27% (27/0)	13% (13/0)	17% (17/0)
2-test	2% (2/0)	2% (2/0)	0% (0/0)
Link	2% (2/0)	2% (2/0)	0% (0/0)
CPM	49% (48/1)	47% (47/0)	38% (37/1)
Leiden	2% (2/0)	1% (1/0)	0% (0/0)
L-CPM	5% (5/0)	2% (2/0)	0% (0/0)
Info.	0% (0/0)	0% (0/0)	0% (0/0)
SLPA	75% (66/9)	20% (5/15)	66% (65/1)
DEMON	14% (12/2)	10% (10/0)	2% (0/2)
ANGEL	15% (14/1)	14% (14/0)	0% (0/0)
SCAN	2% (2/0)	2% (2/0)	0% (0/0)
SCAN-J	19% (18/0)	21% (21/0)	0% (0/0)

TABLE S19. Control families: Ω of HyperCommon (HC), its triangles-only variant and the best other method under the oracle and label-free protocols, with the reference family each control changes (one assumption at a time; Sec. VI), mean over the noise range of each family ($p \in \{0, 0.1, \ldots, 0.8\}$ for the controls, $p \in \{0, \ldots, 0.6\}$ for the swap controls and their references, $p \in [0, 0.8]$ for the other references). The controls are compared with the methods that lead on the main benchmarks (DEMON, ANGEL, SCAN and SCAN-J are left out). Mark on HC as in Table S1.

	oracle			label-free			reference	HC oracle	HC free
Control	HC	tri.	best other	HC	tri.	best other			
hubs, $(50, k_i \in [4, 32]) \times 40$	0.58^{-}	0.53	0.64 (Leiden)	0.56^{+}	0.52	0.53 (Link)	uniform k=8	0.57	0.55
hubs, $(100, k_i \in [8, 64]) \times 20$	0.71^{-}	0.72	0.84 (Leiden)	0.69^{-}	0.70	0.83 (Leiden)	uniform k=16	0.73	0.72
hubs, two worlds	0.76^{-}	0.75	0.85 (L-CPM)	0.75^{-}	0.73	0.83 (L-CPM)	two worlds	0.78	0.76
split overlap, two worlds, o_n=0.05	$0.78^{=}$	0.77	0.78 (L-CPM)	0.78^{+}	0.76	0.71 (Link)	two worlds	0.78	0.76
split overlap, two worlds, o_n=0.10	$0.72^{=}$	0.72	0.73 (L-CPM)	0.72^{+}	0.71	0.62 (CPM)	two worlds, o_n=0.10	0.72	0.71
split overlap, three big, o_n=0.05	0.79^{+}	0.81	0.75 (Link)	0.79^{+}	0.77	0.74 (Link)	three big	0.80	0.78
split overlap, three big, o_n=0.10	0.73^{+}	0.75	0.66 (CPM)	0.72^{+}	0.69	0.62 (Link)	three big, o_n=0.10	0.71	0.67
swaps, uniform k=8, q=0.1	0.68^{-}	0.56	0.81 (Leiden)	$0.65^{=}$	0.56	0.66 (Leiden)	endpoint, q=0.1	0.68	0.65
swaps, uniform k=8, q=0.3	0.63^{-}	0.41	0.85 (Leiden)	0.57^{-}	0.41	0.70 (Info.)	endpoint, q=0.3	0.62	0.57
swaps, three big, q=0.1	0.92^{-}	0.92	0.95 (L-CPM)	0.91^{+}	0.91	0.85 (CPM)	endpoint, q=0.1	0.92	0.91
swaps, three big, q=0.3	0.84^{-}	0.83	0.96 (L-CPM)	0.84^{-}	0.80	0.94 (L-CPM)	endpoint, q=0.3	0.84	0.83
random geometric, $(50, 8) \times 40$	0.48^{-}	0.43	0.57 (Leiden)	$0.46^{=}$	0.43	0.46 (Leiden)	uniform k=8	0.57	0.55
random geometric, two worlds	0.73^{-}	0.74	0.84 (L-CPM)	0.73^{-}	0.73	0.79 (L-CPM)	two worlds	0.78	0.76

TABLE S20. Null graphs with triadic closure but no planted communities: number of communities and coverage of each method's label-free cover (mean over ten graphs). RGG: random geometric graph on the unit torus, 2000 nodes, mean degree 8 (clustering 0.59); HK: Holme–Kim graph, 2000 nodes, $m = 4$, triangle probability 0.9 (clustering 0.43).

Graph		HC	tri.	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	DEMON	ANGEL	SCAN	SCAN-J
RGG	communities	101	98	188	162	34	35	36	53	178	31	172	237	89	92	66
	coverage	0.98	0.98	0.99	0.93	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.98	0.96	0.97
HK	communities	175	166	391	288	16	16	27	32	55	12	42	196	156	97	121
	coverage	0.96	0.85	1.00	0.66	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.76	0.82	0.82

TABLE S21. HyperCommon with each commonality function, best threshold per graph: mean Ω over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level, the threshold a quantile of each function's triple scores. Superscripts mark a paired 95% interval of the difference from Jaccard that excludes zero ($^+$ better, $^-$ worse). Bold: best single function in the row (last column excluded). $I/\max$, $I/\min$: overlap normalised by the larger or smaller neighbourhood; I: overlap count; RA: resource allocation. Last column: function and level chosen together on each graph by the same rule. Quantile levels $0, 0.04, \ldots, 0.96$ and $0.97, 0.98, 0.99, 0.995$; rings sampled every 0.1 in p. Jaccard values can differ from the HC columns of Tables S1 and S3 (by up to 0.04) because thresholds are quantile levels and rings are sampled every 0.1 in p.

Family	Jaccard	cosine	$I/\max$	$I/\min$	I^2/U	I	RA	per graph
uniform, k=8, o_n=0	**0.62**	**0.62**	0.61$^-$	0.61$^-$	**0.62**	0.56$^-$	0.57$^-$	0.63
uniform, k=8, o_n=0.05	**0.58**	**0.58**	0.56$^-$	0.54$^-$	0.57$^-$	0.52$^-$	0.49$^-$	0.59
uniform, k=8, o_n=0.10	0.51	0.51	0.50$^-$	**0.52**	0.50$^-$	0.46$^-$	0.43$^-$	0.53
uniform, k=16, o_n=0	0.78	**0.79**$^+$	0.77$^-$	0.77$^-$	0.78	0.74$^-$	0.74$^-$	0.79
uniform, k=16, o_n=0.05	**0.72**	**0.72**	**0.72**	0.71$^-$	0.70$^-$	0.63$^-$	0.62$^-$	0.74
uniform, k=16, o_n=0.10	**0.64**	0.63$^-$	**0.64**	0.62$^-$	0.58$^-$	0.43$^-$	0.45$^-$	0.67
equal k, unequal s (k=8)	**0.52**	**0.52**	0.51$^-$	0.50$^-$	**0.52**	0.46$^-$	0.44$^-$	0.53
equal k, unequal s (k=16)	**0.74**	**0.74**	**0.74**	0.73	0.73$^-$	0.67$^-$	0.66$^-$	0.76
equal s, unequal k	0.73	0.73$^-$	**0.74**	0.64$^-$	0.65$^-$	0.52$^-$	0.57$^-$	0.75
two worlds	0.80	0.78$^-$	**0.81**	0.66$^-$	0.75$^-$	0.67$^-$	0.66$^-$	0.82
three big, many small	0.81	0.80$^-$	**0.83**$^+$	0.68$^-$	0.77$^-$	0.62$^-$	0.63$^-$	0.85
gradient of ten rings	0.59	0.58$^-$	**0.60**$^+$	0.51$^-$	0.58	0.52$^-$	0.52$^-$	0.65
LFR, o_n=0	0.84	**0.85**$^+$	0.81$^-$	0.84	**0.85**	0.76$^-$	0.72$^-$	0.87
LFR, o_n=0.1	0.73	**0.74**$^+$	0.71$^-$	**0.74**$^+$	0.70$^-$	0.59$^-$	0.53$^-$	0.77
LFR, o_n=0.3	0.49	**0.50**	0.48$^-$	**0.50**	0.43$^-$	0.32$^-$	0.30$^-$	0.53
LFR, $\langle k \rangle$=10	0.47	**0.48**$^+$	0.42$^-$	0.43$^-$	0.47	0.40$^-$	0.26$^-$	0.49
LFR, o_m=4	0.61	0.61	0.60$^-$	**0.62**$^+$	0.58$^-$	0.47$^-$	0.45$^-$	0.63
LFR, sizes 50–250	0.45	**0.46**$^+$	0.41$^-$	**0.46**	**0.46**	0.29$^-$	0.25$^-$	0.51
better / worse than Jaccard	–	5 / 5	2 / 11	2 / 11	0 / 11	0 / 18	0 / 18	

TABLE S22. HyperCommon with each commonality function, label-free threshold (largest overlapping modularity): mean Ω over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level, the threshold a quantile of each function's triple scores. Superscripts mark a paired 95% interval of the difference from Jaccard that excludes zero ($^+$ better, $^-$ worse). Bold: best single function in the row (last column excluded). $I/\max$, $I/\min$: overlap normalised by the larger or smaller neighbourhood; I: overlap count; RA: resource allocation. Last column: function and level chosen together on each graph by the same rule. Quantile levels $0, 0.04, \ldots, 0.96$ and $0.97, 0.98, 0.99, 0.995$; rings sampled every 0.1 in p. Jaccard values can differ from the HC columns of Tables S1 and S3 (by up to 0.04) because thresholds are quantile levels and rings are sampled every 0.1 in p.

Family	Jaccard	cosine	$I/\max$	$I/\min$	I^2/U	I	RA	per graph
uniform, k=8, o_n=0	0.59	0.60$^+$	0.59	**0.61**$^+$	0.58	0.55$^-$	0.54$^-$	0.61
uniform, k=8, o_n=0.05	**0.56**	**0.56**	0.55	0.50$^-$	0.55$^-$	0.51$^-$	0.45$^-$	0.56
uniform, k=8, o_n=0.10	**0.48**	**0.48**	**0.48**	0.47	0.46$^-$	0.44$^-$	0.40$^-$	0.48
uniform, k=16, o_n=0	0.78	**0.79**$^+$	0.77$^-$	0.77$^-$	0.78	0.74$^-$	0.74$^-$	0.79
uniform, k=16, o_n=0.05	**0.71**	**0.71**	**0.71**	0.69$^-$	0.69$^-$	0.62$^-$	0.60$^-$	0.72
uniform, k=16, o_n=0.10	**0.63**	0.62$^-$	0.62	0.60$^-$	0.56$^-$	0.41$^-$	0.42$^-$	0.65
equal k, unequal s (k=8)	0.51	**0.52**	0.50$^-$	0.49$^-$	0.51	0.45$^-$	0.43$^-$	0.52
equal k, unequal s (k=16)	0.72	**0.73**	**0.73**	0.71$^-$	0.70$^-$	0.64$^-$	0.64$^-$	0.73
equal s, unequal k	0.71	0.70$^-$	**0.72**$^+$	0.62$^-$	0.62$^-$	0.48$^-$	0.55$^-$	0.73
two worlds	0.78	0.77$^-$	**0.80**	0.63$^-$	0.71$^-$	0.66$^-$	0.63$^-$	0.81
three big, many small	0.78	0.77	**0.82**$^+$	0.62$^-$	0.63$^-$	0.44$^-$	0.49$^-$	0.81
gradient of ten rings	0.36	0.36	0.39$^+$	0.32	**0.50**$^+$	0.48$^+$	0.42	0.31
LFR, o_n=0	0.81	0.82	0.80	**0.84**$^+$	0.81	0.70$^-$	0.67$^-$	0.82
LFR, o_n=0.1	0.70	**0.72**$^+$	0.67$^-$	0.69	0.65$^-$	0.56$^-$	0.45$^-$	0.73
LFR, o_n=0.3	0.43	0.43	**0.44**	**0.44**	0.37$^-$	0.26$^-$	0.28$^-$	0.45
LFR, $\langle k \rangle$=10	0.41	**0.44**	0.38$^-$	0.37$^-$	0.41	0.40	0.21$^-$	0.43
LFR, o_m=4	0.60	0.60	0.59	**0.61**	0.57$^-$	0.45$^-$	0.41$^-$	0.62
LFR, sizes 50–250	0.43	**0.44**	0.40$^-$	**0.44**	**0.44**	0.28$^-$	0.25$^-$	0.48
better / worse than Jaccard	–	3 / 3	3 / 5	2 / 10	1 / 11	1 / 16	0 / 17	

TABLE S23. As Table III, detecting on the subgraphs induced by the labelled nodes (16 716 nodes for Amazon, 93 432 for DBLP, 47 135 for YouTube). The induced Amazon subgraph falls into 1 106 components that already score F1 0.958, and on YouTube the components are the most accurate cover by F1, so this setting does not discriminate between methods on those two. Clique percolation on YouTube did not complete (memory above 100 GB). Bold: best in column among the methods; the components row is a reference and is not marked.

	Amazon				DBLP				YouTube			
	F1 best	F1 free	Ω best	Ω free	F1 best	F1 free	Ω best	Ω free	F1 best	F1 free	Ω best	Ω free
components	0.958	0.958	0.689	0.689	0.531	0.531	0.003	0.003	0.504	0.504	0.001	0.001
HyperCommon	0.959	0.958	0.697	0.680	0.584	0.584	0.046	0.046	0.477	0.432	0.029	0.029
HC_f	**0.963**	0.958	0.721	**0.689**	**0.614**	0.583	0.073	0.047	**0.504**	0.392	0.031	0.021
edge-connected	0.959	0.958	0.697	0.680	0.584	0.584	0.045	0.044	0.477	0.433	0.030	0.030
triangles only	0.870	0.870	0.414	0.413	0.576	0.549	0.058	0.031	0.426	0.376	0.018	0.018
two-test	0.958	0.958	0.706	0.687	0.569	0.519	0.022	0.022	0.480	0.432	0.011	0.011
link comm.	0.959	0.959	0.698	0.683	0.541	0.539	0.061	0.055	0.476	0.431	0.146	0.024
CPM	0.870	0.870	0.414	0.414	0.573	0.534	0.058	0.058	–	–	–	–
Leiden	0.963	**0.963**	**0.744**	0.652	0.590	0.458	0.202	**0.178**	**0.504**	**0.470**	**0.159**	0.077
L-CPM	0.963	**0.963**	0.689	0.662	0.569	0.448	**0.205**	0.177	0.496	0.236	0.127	0.071
Infomap	0.963	**0.963**	0.637	0.637	0.609	0.552	0.095	0.095	0.502	0.450	0.134	**0.084**
label prop.	0.836	0.836	0.381	0.381	0.552	0.552	0.012	0.012	0.420	0.420	0.037	0.037
SLPA	0.870	0.859	0.421	0.398	0.605	**0.604**	0.024	0.018	0.430	0.430	0.040	0.038

TABLE S24. As Table III, with F1 scored against the distinct node sets of the recorded communities rather than against the top-5000 lists, which repeat some sets.

	Amazon					DBLP					YouTube				
	F1 best	F1 free	Ω best	Ω free	cov.	F1 best	F1 free	Ω best	Ω free	cov.	F1 best	F1 free	Ω best	Ω free	cov.
components	0.020	0.020	0.000	0.000	1.00	0.075	0.075	0.000	0.000	1.00	0.060	0.060	0.000	0.000	1.00
HC	0.848	0.847	0.459	0.459	0.99	0.515	0.486	0.024	0.024	0.94	0.321	0.288	0.017	−0.007	0.67
HC_f	0.874	0.844	0.477	0.477	1.00	0.520	0.486	0.037	0.024	0.94	–	–	–	–	–
e-conn.	0.851	0.851	0.485	0.485	0.99	0.515	0.487	0.024	0.024	0.94	0.322	0.291	0.016	−0.007	0.67
tri.	0.830	0.828	0.415	0.415	0.95	0.513	0.484	0.030	0.019	0.89	0.308	0.299	0.015	0.006	0.49
2-test	0.878	0.878	0.424	0.424	0.95	0.507	0.459	0.014	0.014	0.83	0.279	0.215	0.005	−0.011	0.53
Link	0.838	0.820	0.487	0.487	1.00	0.490	0.446	0.031	0.024	1.00	–	–	–	–	–
Leiden	0.923	0.563	0.600	0.392	1.00	0.542	0.285	**0.154**	**0.152**	1.00	**0.362**	0.192	**0.142**	0.074	1.00
Leiden, γ=1	0.563	0.563	0.392	0.392	1.00	0.285	0.285	0.152	**0.152**	1.00	0.192	0.192	0.074	0.074	1.00
L-CPM	0.927	0.587	**0.617**	0.385	1.00	0.532	0.237	0.150	0.150	1.00	0.337	0.252	0.113	0.104	1.00
Infomap	**0.935**	0.903	0.577	**0.576**	1.00	**0.547**	0.505	0.041	0.030	1.00	0.354	**0.338**	0.128	**0.128**	1.00
Infomap, Markov time 1	0.915	**0.915**	0.488	0.488	1.00	**0.547**	**0.547**	0.013	0.013	1.00	0.338	**0.338**	0.128	**0.128**	1.00
LPA	0.807	0.807	0.373	0.373	1.00	0.507	0.507	0.010	0.010	1.00	0.217	0.217	0.007	0.007	1.00

TABLE S25. Every method on the fourteen smaller networks: best-match F1 at the best parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J
College football	0.91	0.91	0.91	**0.91**	0.68	0.85	0.89	0.89	0.89	0.83	0.87	0.77	0.81	0.75	0.89	0.89
Political books	**0.78**	0.74	0.66	0.78	0.71	0.65	**0.79**	**0.79**	0.77	0.62	0.65	0.71	0.75	0.75	0.77	0.66
Political blogs	**0.64**	**0.64**	0.35	**0.64**	**0.64**	–	0.43	0.37	0.53	0.43	0.42	0.57	0.60	0.59	0.62	0.54
Primary school	0.40	0.40	0.44	0.42	0.52	–	**0.89**	**0.89**	0.19	0.19	0.55	0.19	0.68	0.64	0.51	0.46
Facebook ego 0	0.40	0.40	0.40	**0.41**	0.38	0.40	0.38	0.36	0.40	0.36	0.34	0.39	**0.42**	0.39	**0.42**	0.41
Facebook ego 107	0.55	0.53	0.52	0.60	0.48	–	0.62	0.59	0.69	0.64	0.62	**0.71**	0.59	0.55	0.63	0.62
Facebook ego 348	0.65	0.65	0.63	0.66	0.61	0.66	0.66	0.66	0.66	0.51	0.46	**0.67**	0.61	0.66	0.66	0.65
Facebook ego 414	0.73	0.73	**0.76**	0.75	0.62	0.75	**0.76**	**0.76**	**0.76**	**0.76**	0.67	**0.76**	0.70	0.73	**0.76**	0.75
Facebook ego 686	0.58	0.58	**0.68**	0.62	0.55	0.55	0.61	0.55	0.61	0.55	0.42	0.55	0.62	0.63	0.62	0.61
Facebook ego 698	0.83	0.83	0.83	0.82	0.76	**0.85**	0.81	0.81	0.82	0.81	0.77	0.80	0.82	0.76	0.82	0.83
Facebook ego 1684	0.61	0.61	0.59	0.63	0.42	–	0.62	0.56	**0.64**	0.57	0.55	0.62	0.54	0.57	0.63	0.62
Facebook ego 1912	0.46	0.46	0.41	0.45	0.42	–	**0.53**	0.45	0.50	0.47	0.47	0.52	0.49	0.48	0.44	0.44
Facebook ego 3437	0.56	0.56	0.56	0.61	0.52	0.51	**0.70**	**0.70**	0.67	0.54	0.56	0.54	0.62	0.59	0.61	0.61
Facebook ego 3980	0.68	0.68	0.68	0.68	0.68	0.61	0.71	0.68	0.68	**0.73**	0.62	0.66	0.63	0.63	0.68	0.68
mean rank	7.3	7.9	8.6	5.9	12.4	10.1	5.5	8.0	6.3	10.7	12.3	8.6	8.1	9.3	5.4	7.6

TABLE S26. Every method on the fourteen smaller networks: best-match F1 at the label-free parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J
College football	0.88	0.88	0.86	**0.89**	0.42	0.85	0.87	0.83	0.88	0.83	0.87	0.77	0.69	0.74	**0.89**	**0.89**
Political books	0.71	0.70	0.59	**0.78**	0.71	0.65	0.66	0.72	0.66	0.62	0.65	0.69	0.71	0.62	0.77	0.66
Political blogs	0.20	0.19	0.18	0.18	0.29	–	0.42	0.37	0.50	0.43	0.42	**0.57**	0.43	0.49	0.19	0.12
Primary school	0.40	0.40	0.44	0.40	0.31	–	0.77	**0.89**	0.19	0.19	0.55	0.19	0.35	0.64	0.48	0.46
Facebook ego 0	0.39	0.38	0.38	0.39	0.35	0.38	0.37	0.35	0.38	0.36	0.34	0.38	0.40	0.35	0.40	**0.41**
Facebook ego 107	0.55	0.53	0.52	0.52	0.35	–	0.60	0.55	0.63	0.64	0.62	**0.70**	0.59	0.45	0.44	0.48
Facebook ego 348	0.48	0.48	0.48	0.53	0.50	0.47	0.53	0.46	0.58	0.51	0.46	**0.64**	0.58	0.56	0.51	0.51
Facebook ego 414	0.68	0.68	**0.75**	0.70	0.58	0.75	0.67	**0.76**	0.69	**0.76**	0.67	**0.76**	0.70	0.73	0.70	0.70
Facebook ego 686	0.56	0.56	0.58	0.61	0.42	0.46	0.54	0.37	0.30	0.55	0.42	0.54	**0.62**	0.50	**0.62**	0.61
Facebook ego 698	0.83	0.83	0.83	0.82	0.76	**0.85**	0.73	0.81	0.72	0.81	0.77	0.80	0.82	0.73	0.81	0.82
Facebook ego 1684	0.60	0.59	0.58	0.60	0.26	–	0.61	0.50	**0.64**	0.57	0.55	0.61	0.52	0.57	0.62	0.62
Facebook ego 1912	0.37	0.37	0.36	0.42	0.32	–	**0.53**	0.40	0.49	0.47	0.47	0.51	0.44	0.36	0.38	0.38
Facebook ego 3437	0.53	0.53	0.53	0.53	0.45	0.51	**0.64**	0.56	0.56	0.54	0.56	0.54	0.56	0.52	0.57	0.55
Facebook ego 3980	0.57	0.57	0.57	0.50	0.60	0.61	0.68	0.53	0.61	**0.73**	0.62	0.66	0.57	0.50	0.51	0.50
mean rank	8.3	8.9	9.4	7.6	12.5	9.2	6.8	8.8	7.3	8.0	9.3	6.0	6.5	10.8	6.6	7.5

TABLE S27. Every method on the fourteen smaller networks: Ω at the best parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J
College football	0.92	0.92	0.92	**0.93**	0.75	0.84	0.90	0.90	0.88	0.75	0.82	0.63	0.42	0.56	0.85	0.85
Political books	0.69	**0.70**	0.66	0.65	0.65	0.60	0.67	0.68	0.67	0.59	0.65	0.69	0.41	0.65	0.60	0.58
Political blogs	0.04	0.04	0.03	0.03	0.06	–	**0.55**	0.32	0.54	0.53	0.51	0.54	0.10	0.15	0.04	0.01
Primary school	0.06	0.06	0.24	0.06	0.32	–	**0.81**	**0.81**	0.00	0.00	0.38	0.00	0.00	0.00	0.36	0.23
Facebook ego 0	0.29	0.29	0.29	0.27	0.29	0.30	0.31	0.31	**0.34**	0.29	0.26	0.33	0.14	0.29	0.29	0.25
Facebook ego 107	0.51	0.48	0.47	0.53	0.48	–	0.75	**0.84**	0.74	0.74	0.53	0.73	0.23	0.55	0.40	0.41
Facebook ego 348	0.16	0.16	0.13	0.15	0.16	0.13	0.18	0.18	0.18	0.01	−0.09	0.05	**0.29**	0.16	0.16	0.14
Facebook ego 414	0.86	0.86	0.85	0.85	0.80	0.83	0.84	**0.87**	0.83	0.81	0.83	0.82	0.57	0.72	0.85	0.80
Facebook ego 686	**0.11**	**0.11**	**0.11**	0.10	**0.11**	0.10	0.05	**0.11**	0.04	0.00	0.02	−0.01	0.09	0.09	0.09	0.09
Facebook ego 698	0.59	0.59	0.59	0.58	**0.69**	0.58	0.66	0.61	0.62	0.61	0.60	0.58	0.64	0.61	0.55	0.58
Facebook ego 1684	0.74	0.75	0.74	0.75	0.61	–	0.68	0.69	**0.81**	0.40	0.67	0.48	0.31	0.39	0.73	0.73
Facebook ego 1912	0.58	0.58	0.58	0.58	0.35	–	0.64	**0.67**	0.66	0.61	0.60	0.61	0.27	0.35	0.58	0.58
Facebook ego 3437	0.46	0.46	0.46	0.37	0.46	0.47	0.47	0.49	0.46	0.39	**0.50**	0.39	0.26	0.49	0.27	0.34
Facebook ego 3980	0.19	0.19	0.17	0.19	0.19	0.10	0.42	0.21	0.33	**0.53**	0.28	0.17	0.04	0.02	0.19	0.19
mean rank	7.2	7.1	8.3	8.4	8.8	10.1	4.4	3.6	5.5	9.6	8.4	9.6	12.5	9.8	9.6	11.1

TABLE S28. Every method on the fourteen smaller networks: Ω at the label-free parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J
College football	0.84	0.84	**0.89**	0.85	0.24	0.84	0.81	0.71	0.85	0.75	0.82	0.63	0.42	0.56	0.85	0.85
Political books	0.66	**0.68**	0.65	0.65	0.61	0.60	0.59	**0.68**	0.66	0.59	0.65	**0.68**	0.41	0.64	0.60	0.58
Political blogs	0.04	0.04	0.03	0.03	0.00	–	0.52	0.32	0.53	0.53	0.51	**0.54**	0.01	0.15	0.04	0.01
Primary school	0.06	0.06	0.24	0.06	−0.01	–	0.62	**0.81**	0.00	0.00	0.38	0.00	0.00	0.00	0.36	0.23
Facebook ego 0	0.25	0.29	0.29	0.27	0.27	0.25	0.16	0.27	**0.33**	0.29	0.26	0.32	0.12	0.28	0.17	0.22
Facebook ego 107	0.51	0.48	0.47	0.41	0.41	–	0.51	0.54	0.58	**0.74**	0.53	0.73	0.19	0.55	0.40	0.41
Facebook ego 348	−0.08	−0.08	−0.08	−0.08	0.00	−0.07	−0.04	−0.10	−0.10	0.01	−0.09	0.04	**0.14**	0.02	−0.07	−0.07
Facebook ego 414	**0.86**	**0.86**	0.82	0.85	0.80	0.82	0.83	0.81	0.83	0.81	0.83	0.81	0.57	0.72	0.85	0.80
Facebook ego 686	0.08	0.08	0.08	0.07	**0.09**	0.08	0.04	0.04	0.01	0.00	0.02	−0.02	0.09	0.06	0.06	**0.09**
Facebook ego 698	0.59	0.59	0.59	0.58	0.55	0.58	**0.65**	0.61	0.59	0.61	0.60	0.58	**0.64**	0.61	0.55	0.58
Facebook ego 1684	0.74	0.75	0.74	0.73	0.34	–	0.63	0.69	**0.81**	0.40	0.67	0.48	0.26	0.37	0.73	0.73
Facebook ego 1912	0.57	0.57	0.57	0.58	0.27	–	0.60	0.60	**0.66**	0.61	0.60	0.60	0.27	0.33	0.58	0.58
Facebook ego 3437	0.46	0.46	0.44	0.26	0.46	0.47	0.41	0.48	0.45	0.39	**0.50**	0.36	0.26	0.49	0.21	0.19
Facebook ego 3980	0.04	0.04	0.06	0.04	0.11	0.06	0.40	0.20	0.29	**0.53**	0.28	0.17	0.01	−0.05	0.03	0.04
mean rank	7.6	7.0	7.4	8.9	11.3	8.3	7.5	7.0	5.9	7.5	6.9	7.6	12.1	8.9	9.4	9.9

TABLE S29. Every method on the fourteen smaller networks: coverage at the label-free parameter (stochastic methods: mean over five seeds).

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J
College football	1.00	1.00	1.00	0.94	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.94	0.94
Political books	0.95	0.96	0.95	0.97	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.97	0.94	0.88
Political blogs	0.23	0.23	0.21	0.22	0.78	–	1.00	1.00	0.82	1.00	1.00	0.82	0.67	0.63	0.26	0.13
Primary school	0.93	0.93	0.66	0.91	1.00	–	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.59	0.80
Facebook ego 0	0.83	0.87	0.80	0.86	0.97	0.72	1.00	1.00	1.00	1.00	1.00	1.00	0.84	0.83	0.76	0.75
Facebook ego 107	0.88	0.88	0.87	0.80	0.97	–	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.97	0.84	0.86
Facebook ego 348	0.74	0.74	0.74	0.79	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.94	0.71	0.72
Facebook ego 414	0.97	0.97	0.93	0.97	0.99	0.91	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.97	0.99	0.93
Facebook ego 686	0.92	0.92	0.90	0.90	0.96	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.95	0.91	0.93	0.85
Facebook ego 698	0.98	0.98	0.98	0.98	1.00	0.88	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.90	0.92	0.98
Facebook ego 1684	0.81	0.81	0.80	0.81	0.98	–	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.97	0.78	0.77
Facebook ego 1912	0.82	0.82	0.81	0.83	1.00	–	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.97	0.79	0.81
Facebook ego 3437	0.95	0.95	0.94	0.87	0.99	0.90	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.81	0.82
Facebook ego 3980	0.78	0.78	0.78	0.80	0.85	0.68	1.00	1.00	1.00	1.00	1.00	1.00	0.78	0.72	0.78	0.80

[1] D. Hric, R. K. Darst, and S. Fortunato, Community detection in networks: Structural communities versus ground truth, Physical Review E **90**, 062805 (2014).

[2] J. Yang and J. Leskovec, Defining and evaluating network communities based on ground-truth, Knowledge and Information Systems **42**, 181 (2015).

[3] P. Jaccard, The distribution of the flora in the alpine zone, New Phytologist **11**, 37 (1912).

[4] D. Deinega, *Overlapping Communities Detection in Complex Networks*, Master's thesis, Ariel University (2023).

[5] G. Salton and M. J. McGill, *Introduction to Modern Information Retrieval* (McGraw-Hill, 1983).

[6] X. Xu, N. Yuruk, Z. Feng, and T. A. J. Schweiger, SCAN: A structural clustering algorithm for networks, in *Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2007) pp. 824–833.

[7] T. Zhou, L. Lü, and Y.-C. Zhang, Predicting missing links via local information, The European Physical Journal B **71**, 623 (2009).

[8] L. Lü and T. Zhou, Link prediction in complex networks: A survey, Physica A: Statistical Mechanics and its Applications **390**, 1150 (2011).

[9] E. Ravasz, A. L. Somera, D. A. Mongru, Z. N. Oltvai, and A.-L. Barabási, Hierarchical organization of modularity in metabolic networks, Science **297**, 1551 (2002).

[10] G. Palla, I. Derényi, I. Farkas, and T. Vicsek, Uncovering the overlapping community structure of complex networks in nature and society, Nature **435**, 814 (2005).

[11] J. M. Kumpula, M. Kivelä, K. Kaski, and J. Saramäki, Sequential algorithm for fast clique percolation, Physical Review E **78**, 026109 (2008).

TABLE S30. Numbers quoted in the text that are not read directly off a table, and where they are produced (scripts under `paper/`, data in `paper/figures/data/`).

Quantity	Source
Recovery windows of Sec. V and Appendix C (Eq. 5, mixed families, emptiness, three-ring and m-ring windows, separation and small-ring conditions), exact rational arithmetic	`analysis/exact_windows.py` (every check passes)
Shares of rings in a shared pair and fused just above the lower window edge (6%–61% at $o_n = 0.05$)	`analysis/double_bonds.py`; `double_bonds.csv`
Curve values of Fig. 3 (0.92–0.99; 0.39, 0.89, 0.95)	`analysis/fig_omega_t.py`
Label-free against best threshold (27 and 31 of 36 settings); counts of settings by paired mark for Ω, ONMI and all-community recovery under the four protocols; ablation and link-community counts	`analysis/numbers3.py`; `numbers3.csv`, `scorecard3_counts.csv`
Selector on the gradient (113 of 170 graphs above the oracle threshold; 181 against 147 communities)	`analysis/make_tables3.py`; Tables S13 and S15
Isolated-merge recovery ($\Omega = 1$; 1.000 and 0.991 at $p = 0.2$; pair rule 0.74)	`bench/results2/rings_isolated.csv`, best threshold per graph
The gradient at $p = 0$ (Leiden 0.28, and 0.26 at about ten communities; HyperCommon 0.99 and 0.97)	`bench/results2/rings_hetero.csv`, family `gradient10`, $p = 0$
Per-community recovery of large and small rings on the gradient, oracle values to three decimals, and the other family values quoted from the summaries	`bench/results2/<exp>_summary2.csv` (metrics `omega`, `rec_small`, `rec_large`)
LFR at $\mu = 0.1$ (Fig. S1(f))	`analysis/fig_benchmarks2.py`; `benchmarks_curves2.csv`
Geometry values of Sec. VII E (communities, coverage, per-community recovery, ONMI, triangles per edge)	`bench/results2/rings_geometry_summary2.csv`; `analysis/make_tables2.py geometry`
Pair against triple rule on the real networks (base rates, precision at matched recall, pure triples)	`analysis/real_networks_v2.py`; `analysis/out/<net>_v2.json`
Label-free covers of the fourteen smaller networks (e.g. the number of communities on political blogs)	`bench/real_more_detection_v3.py`; `bench/results/real_more_v3.csv`
Native against label-free choices (communities and Ω of link communities and of the Constant Potts Model)	`bench/results2/<exp>_summary2.csv` (protocols `native`, `label_free`)
Controls and null graphs	`analysis/make_tables3_controls.py`; `controls3.csv`
Commonality signal on LFR graphs (AUC 0.97; passing wedges by community size)	`analysis/lfr_signal.py`; `lfr_signal.csv`
Triangles across LFR communities (2%, 19%, 32%)	`analysis/lfr_triangles.py`; `lfr_triangles.csv`
Label-free retention of F1 on real networks (median 94%)	`analysis/derived_numbers.py`; `real_label_free_retention.csv`
Degree-ratio pruning: identical covers at all 30 thresholds on Amazon and DBLP	`bench/check_pruning.py`; `bench/results/check_pruning.csv`
Running times and fitted exponents (Fig. 5)	`bench/runtime.py`, `analysis/fig_runtime.py`; `runtime_summary.csv`, `runtime_exponents.csv`
Single-thread re-timing on Amazon and DBLP	`bench/retime_big.py`; `bench/results/retime_big.csv`
YouTube wedges and pruning share	`analysis/youtube_wedges.py`; `youtube_wedges.csv`
SBM graphs identical to those of the other methods (edge counts of every graph)	`bench/run_sbm.py` rows against `bench/run2.py` rows